AERODYNAMICS PERFORMANCE ANALYSIS OF HYPERSONIC VEHICLES

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BACHELOR OF TECHNOLOGY

In

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By

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Lovely Professional University Jalandhar, Punjab

CERTIFICATE

I hereby certify that the work which is being presented in the capstone entitled **“Aerodynamics Performance Analysis of Hypersonic Vehicles”** in partial fulfillment of the requirement for the award of degree of Bachelor of Technology and submitted in Department of Aerospace Engineering, Lovely Professional University, Punjab is an authentic record of my own work carried out during the period of minor project under the supervision of Dr. Rahul Kumar, Assistant Professor, Department of Aerospace Engineering, Lovely Professional University, Punjab.

The matter presented in this minor has not been submitted by me anywhere for the award of any other degree or to any other institute.

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DECLARATION

We hereby declare that the research work reported in the dissertation entitled *“Aerodynamics Performance Analysis of Hypersonic Vehicles”* submitted in partial fulfilment of the requirements for the award of the Degree of Bachelor of Technology in Aerospace Engineering at Lovely Professional University, Phagwara, Punjab, is an authentic work carried out under the supervision of our research supervisor, Dr. Rahul Kumar. We have not submitted this work elsewhere for the award of any other degree or diploma.

We understand that the work presented herein is in direct compliance with Lovely Professional University’s policies on plagiarism, intellectual property rights, and the highest standards of moral and ethical conduct. Therefore, to the best of our knowledge, the content of this dissertation represents an authentic and honest research effort conducted entirely by us. We are fully responsible for the contents of this dissertation work.

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ABSTRACT

Hypersonic vehicles operating at velocities exceeding Mach 5 encounter a distinct set of aerodynamic challenges arising from shock-wave interactions, extreme aerodynamic heating, compressibility effects, and complex boundary-layer behaviour. The nose geometry of a hypersonic aircraft has a profound influence on the distribution of pressure, temperature, density, and the resultant aerodynamic forces—particularly drag and lift. A thorough numerical investigation of these effects is essential both for fundamental understanding and for informing the design of next-generation high-speed aerial platforms.

This study presents a comprehensive computational fluid dynamics (CFD) analysis of three distinct hypersonic aircraft configurations, each characterised by a different nose geometry. Model 1 is inspired by the NASA X-43A (Hyper-X) and features a flat, rectangular forebody cross-section with a blunt leading edge. Model 2 adopts a moderately elongated conical nose, while Model 3 incorporates a sharply elongated conical nose of the type commonly associated with conventional hypersonic cruise missiles and research vehicles. The total lengths of the three models are 11,142.03 mm, 12,164.23 mm, and 12,843.03 mm, respectively.

All three configurations were simulated at freestream Mach numbers of 5 and 7 using ANSYS Fluent 2024 R1, employing a three-dimensional density-based implicit solver with the SST k-omega turbulence model. The computational domain was discretised using an unstructured tetrahedral mesh, and the ideal-gas law with Sutherland viscosity was applied to capture compressibility effects. Boundary conditions consistent with hypersonic freestream conditions at an inlet temperature of 300 K were imposed.

Results include scaled residual plots confirming solution convergence, time-history plots of the coefficient of drag (Cd), coefficient of lift (Cl), drag force, and lift force, as well as contour plots of the Mach number, static pressure, static temperature, and density distributions over each model. Key findings indicate that nose bluntness significantly elevates bow-shock stand-off distance, stagnation pressure, and peak surface temperature, while nose sharpness reduces wave drag at the cost of concentrated heating at the nose tip. The elongated conical Model 3 demonstrates the most favourable Cd at Mach 7, with drag coefficients decreasing as freestream Mach number rises due to the changed shock structure.

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LIST OF ABBREVIATIONS

Abbreviations	Full Form
CFD	Computational Fluid Dynamics
Cd	Coefficient of Drag
Cl	Coefficient of Lift
SST	Shear Stress Transport
RANS	Reynolds-Averaged Navier-Stokes
TPS	Thermal Protection System
DNS	Direct Numerical Simulation
LES	Large Eddy Simulation
AoA	Angle of Attack
EMU	English Mass Unit
NASA	National Aeronautics and Space Administration
X-43A	NASA Hyper-X Research Aircraft
Ma / M	Mach Number
Pa	Pascal (unit of pressure)
K	Kelvin (unit of temperature)
kg/m^3	Kilogram per cubic metre (density)
W/(m·K)	Watts per metre-Kelvin (thermal conductivity)
J/(kg·K)	Joules per kilogram-Kelvin (specific heat)
DXA	Document Exchange Absolute units (1440 DXA = 1 inch)
AMG	Algebraic Multigrid
CFL	Courant-Friedrichs-Lewy number

CHAPTER 1

INTRODUCTION

1.1 Background of the Study

The pursuit of sustained hypersonic flight has been one of the most challenging frontiers in aerospace engineering for over six decades [1]. A vehicle is classified as hypersonic when it travels at speeds exceeding Mach 5—roughly five times the speed of sound, or approximately 1,700 metres per second under standard sea-level conditions. At such velocities, the physical behaviour of the surrounding airflow departs dramatically from the familiar subsonic and supersonic regimes. Strong bow shocks form ahead of the vehicle, compressing the air to extreme temperatures; the thin boundary layer becomes thermally critical; chemical dissociation of air molecules becomes possible; and real-gas effects begin to influence the flow field in ways that cannot be captured by simple ideal-gas assumptions [2].

Interest in hypersonic flight has grown significantly in the contemporary era for a range of reasons. Hypersonic cruise missiles and glide vehicles now form a central component of emerging defence strategies across multiple nations. At the same time, commercial and governmental interests in affordable, rapid access to low Earth orbit have revived efforts towards reusable hypersonic transport concepts. Space agencies continue to rely on hypersonic re-entry dynamics for the safe return of crewed and uncrewed vehicles [7]. In each of these applications, a thorough understanding of the aerodynamic performance—particularly the distribution of aerodynamic forces and surface heating—is indispensable at the design stage. The nose geometry of a hypersonic vehicle plays a disproportionately large role in determining its aerodynamic character. A blunt nose, such as that seen on re-entry capsules and on the NASA X-43A scramjet demonstrator, generates a detached bow shock that stands off from the vehicle surface. This stand-off distance distributes the aerodynamic heating over a broader surface area, making blunt configurations thermally more survivable at high altitudes. However, the same geometry increases wave drag significantly. Conical or sharply pointed nose shapes, on the other hand, produce weaker oblique shocks that remain attached at the tip, yielding lower wave drag but concentrating heating at the stagnation point [4], [9]. The trade-off between drag and heating is therefore a central design challenge in hypersonic aerodynamics. Computational Fluid Dynamics has become the primary investigative tool for hypersonic aerodynamic studies. Experimental testing at true hypersonic Mach numbers and altitudes is extraordinarily expensive and technically difficult; wind tunnel facilities capable of sustained Mach 7 flows are rare and costly to operate. High-fidelity CFD simulations using commercially mature solvers such as ANSYS Fluent allow researchers to reproduce the complex shock structures, boundary-layer behaviour, and heat-transfer characteristics of hypersonic flows at a fraction of the cost and with a degree of spatial and temporal resolution unattainable in physical experiments [8], [15].

1.2 Problem Statement

The aerodynamic performance—specifically the drag coefficient, lift coefficient, and the spatial distribution of pressure, temperature, and density—of a hypersonic aircraft is critically sensitive to the shape of its nose geometry. There is, however, a lack of systematic comparative studies that place multiple nose configurations side by side under identical freestream conditions, using a consistent numerical framework, at more than one Mach number. Without such comparisons it is difficult to quantify the extent to which nose geometry alone influences the aerodynamic figures of merit that govern the feasibility of a hypersonic platform.

This project addresses that gap by simulating three hypersonic aircraft models—each with a distinct nose geometry, spanning the range from the flat rectangular cross-section of a scramjet-based vehicle to progressively sharpened conical configurations—at freestream Mach numbers of 5 and 7. The primary objective is to document and explain the differences in aerodynamic behaviour that arise from nose-shape variation, and to identify which configuration offers the most aerodynamically favourable performance at hypersonic speeds.

1.3 Scope of the Study

The scope of the present study is defined as follows:

• Three hypersonic aircraft models differing in nose geometry are considered: a blunt flat-fronted configuration (Model 1, inspired by NASA X-43A), a moderate conical nose (Model 2), and an elongated sharp conical nose (Model 3).

• All simulations are performed at freestream Mach numbers of 5 and 7, yielding six distinct simulation cases.

• The flow is treated as three-dimensional, steady, compressible, and turbulent, with the SST k-omega two-equation turbulence model employed in all cases [10].

• The working fluid is air modelled as an ideal gas with Sutherland viscosity [15]. Chemical reactions and high-temperature gas effects beyond compressibility are not considered.

• Post-processing includes convergence monitoring (scaled residual plots), aerodynamic force and coefficient history plots, and contour visualisations of Mach number, pressure, temperature, and density. The study is purely computational; no physical experimental work is carried out.

1.4 Significance of the Study

The results of this investigation contribute to the growing body of knowledge on hypersonic aerodynamics in several ways. First, the direct numerical comparison of three nose geometries under identical conditions provides a clear, quantified measure of the influence of nose shape on drag, lift, and surface conditions. Second, the study demonstrates and validates a reproducible CFD workflow for hypersonic simulation in ANSYS Fluent, which can be adopted and extended by future researchers. Third, by covering two Mach numbers, the work captures how the aerodynamic hierarchy among configurations shifts as speed increases—a subtlety that single-Mach studies miss.

CHAPTER 2

REVIEW OF LITERATURE

2.1 Introduction to the Review

A substantial body of literature has accumulated over the past several decades addressing the computational and experimental investigation of hypersonic aerodynamics. The following sections summarise the most relevant published work, organising contributions thematically around key topics: fundamental hypersonic flow physics, the role of nose geometry, turbulence modelling for hypersonic flows, and the specific capabilities of ANSYS Fluent in the hypersonic regime.

2.2 Fundamentals of Hypersonic Flow

The classical treatment of hypersonic aerodynamics establishes the theoretical underpinnings of Mach-number similarity, Newtonian impact theory, and thin-shock-layer approximations for blunt and slender bodies. Five dominant phenomena that distinguish hypersonic flight from lower-speed regimes: thin shock layers, entropy-layer effects, viscous interaction, high-temperature gas effects, and low-density effects. The present study operates in a regime—Mach 5 to 7 at standard temperature and density—where the first three of these phenomena are dominant, while real-gas chemistry remains secondary.

The reviews of hypersonic aerothermodynamics from the early ballistic-missile era through modern scramjet programmes, concluding that CFD-based predictions have become sufficiently mature for engineering-level design work, provided that the turbulence model and thermodynamic assumptions are selected carefully. They specifically highlighted the SST k-omega model as well suited to shock-induced separation problems. Numerical investigations of blunt-body heating for planetary re-entry vehicles, demonstrating that nose bluntness ratio—defined as the ratio of nose radius to body diameter—is the single most influential geometric parameter governing stagnation-point heat flux [7]. Validated computational predictions show that a 20% increase in nose radius reduced peak heating by approximately 15%, although this is accompanied by a comparable increase in wave drag. RANS simulations of slender hypersonic bodies at Mach 6 and Mach 8, concluding that sharply conical noses consistently yield drag coefficients 30 to 50 percent lower than blunt-nosed counterparts of identical fineness ratio [4]. However, these configurations exhibit significantly increase in heat-flux concentration at the cone apex for the sharpest geometries, suggesting that combined aerodynamic and thermal constraints strongly favour intermediate nose shapes for practical vehicles. A DLR-TAU code to simulate flow over a series of blunted cone-cylinder bodies at Mach 6, showing that even moderate blunting substantially modified the shock structure and the extent of the entropy layer, with consequent effects on downstream boundary-layer transition [9]. This entropy-layer effect is one mechanism by which the aerodynamic influence of nose geometry propagates far downstream of the nose itself.

2.3 The NASA X-43A (Hyper-X) Programme

The NASA X-43A aircraft, which achieved Mach 9.6 in 2004, represents the most closely analogous configuration to Model 1 in the present study. Pre-flight CFD predictions for the X-43A aerodynamic database, noting that the vehicle's flat-bottomed, blunt-nosed configuration was driven primarily by the need to accommodate the scramjet inlet under the forebody and to use the shock compression of the forebody as part of the engine cycle [3]. The CFD predictions showed good agreement with wind-tunnel data for lift, drag, and pitching moment across Mach 3 to 10. Further aerodynamic characterisation of the full X-43A stack, highlight the strong coupling between the forebody shock, the scramjet inlet flow, and the aft body nozzle expansion [5]. While the present study does not model the propulsion system, the flow structure around the Model 1 forebody is expected to exhibit analogous features—in particular, the strong bow shock ahead of the flat leading edge and the associated high pressure and temperature on the lower surface.

2.4 Turbulence Modelling for Hypersonic Flows

The choice of turbulence model is a critical decision in any hypersonic CFD simulation. The SST k-omega model is formulated as a hybrid that blends the k-omega formulation in the near-wall region—where it performs well in adverse pressure gradients—with the k-epsilon formulation in the freestream, thereby avoiding the sensitivity of pure k-omega to freestream turbulence levels [10]. It has been confirmed that the SST model provides consistently more accurate predictions of shock-induced separation than either of its parent models for a range of high-speed configurations [6]. Validation studies on turbulence models for hypersonic flows indicate that no single two-equation model performs uniformly well across all flow features. However, the SST k-omega model emerged as the best overall performer for attached and mildly separated hypersonic boundary layers—precisely the conditions encountered in the present simulations. The commonly recommended inlet conditions of 5% turbulent intensity and a turbulent viscosity ratio of 10 are adopted in this study [11].

2.5 ANSYS Fluent in Hypersonic Simulation

The density-based implicit solver in ANSYS Fluent has been validated against experimental data for Mach 5 flow over a blunted double-cone configuration, demonstrating that the solver captured the complex shock–shock interaction and the resulting localised heating with engineering accuracy when a second-order upwind scheme was used and the mesh was sufficiently refined in regions of high gradients. Their work supports the solver settings adopted in the present study. Similarly, simulations for Mach 7 flow over a hollow cylinder–flare geometry, confirming that the SST k-omega model in ANSYS Fluent correctly predicted the separation length and reattachment heat flux to within 10% of measured values when y+ was maintained below 1 in the near-wall region [12].

2.6 Research Gap

While the published literature comprehensively addresses individual aspects of hypersonic aerodynamics—blunt-body heating, conical wave drag, turbulence model selection, and solver validation—there is a relative scarcity of comparative studies that systematically vary nose geometry across more than two configurations under identical numerical conditions at multiple Mach numbers [4]. Most comparative studies in the open literature involve either two-dimensional axisymmetric geometries or limit themselves to a single Mach number. The present study addresses this gap by performing fully three-dimensional, multi-Mach-number simulations of three geometrically distinct configurations, producing a comparative dataset that can inform practical design choices.

2.7 Summary

The reviewed literature establishes a strong consensus on several points that inform the present study: the SST k-omega model is appropriate for the flow regime of interest; the density-based implicit solver in ANSYS Fluent is capable of capturing hypersonic shock structures with engineering accuracy; nose geometry is the dominant geometric variable governing both wave drag and stagnation-point heating; and direct numerical comparison of multiple nose shapes under consistent conditions represents a meaningful contribution to the field.

CHAPTER 3

RATIONALE AND SCOPE OF THE STUDY

3.1 Rationale

The aerodynamic optimisation of hypersonic vehicles is a problem of growing strategic and commercial importance. Across the globe, investments in hypersonic cruise missiles, boost-glide vehicles, scramjet-powered aircraft, and reusable space-access vehicles have intensified, driving a corresponding demand for reliable, cost-effective aerodynamic analysis tools and validated parametric datasets [2]. The nose shape of a hypersonic vehicle is among the most consequential design variables available to an aerodynamicist. It determines the type, strength, and stand-off distance of the leading bow shock; the distribution of stagnation pressure and temperature along the forebody; the extent of entropy-layer swallowing; and the overall wave-drag contribution to the total drag budget [7]. Despite this acknowledged importance, systematic multi-model, multi-Mach comparative CFD studies are rare in the open literature—particularly those that include a vehicle geometry as unconventional as the flat-nosed X-43A-type configuration alongside traditional conical bodies. The rationale for the present study therefore rests on three pillars. First, there is a clear scientific need for a controlled, consistent numerical comparison of aerodynamic performance across nose shapes. Second, the availability of ANSYS Fluent 2024 R1 with validated settings for hypersonic simulation makes such a comparison feasible within the timescale and resource constraints of a capstone project [15], [16]. Third, the results have direct educational value, providing the student researchers with deep practical experience in high-speed CFD methodology.

3.2 Scope

The scope of the present study encompasses the following:

• Three original hypersonic aircraft geometries, designed in CAD and meshed in ANSYS Meshing, each representing a different nose shape philosophy.

• Simulation at two freestream Mach numbers—Mach 5 and Mach 7—representing entry-level and deep-hypersonic conditions respectively.

• A fully three-dimensional, steady, compressible, turbulent flow analysis using ANSYS Fluent with SST k-omega and a density-based implicit solver.

• Post-processing comprising residual plots, aerodynamic coefficient and force history plots, and five types of flow-field contour maps.

• A comparative analysis of results across all models and Mach numbers, with physical explanation of observed trends based on hypersonic aerodynamic theory [4].

3.3 Problem Statement

Given three hypersonic aircraft configurations with distinct nose geometries—a blunt flat-rectangular nose (Model 1, total length 11,142.03 mm), a moderate conical nose (Model 2, total length 12,164.23 mm), and an elongated sharp conical nose (Model 3, total length 12,843.03 mm)—how do the aerodynamic performance metrics (coefficient of drag, coefficient of lift, drag force, lift force) and the flow-field characteristics (Mach number distribution, static pressure, static temperature, density) differ when each configuration is subjected to hypersonic freestream conditions at Mach 5 and Mach 7. Which nose geometry offers the most favourable aerodynamic performance for hypersonic cruise applications based on established aerodynamic principles.

CHAPTER 4

OBJECTIVES AND HYPOTHESIS

4.1 Objectives

The primary and secondary objectives of this capstone project are listed below.

4.1.1 Primary Objectives

- To perform three-dimensional CFD simulations of three hypersonic aircraft configurations with distinct nose geometries at Mach 5 and Mach 7 using ANSYS Fluent 2024 R1 [15].
- To quantify and compare the aerodynamic performance metrics—specifically the coefficient of drag (CD), coefficient of lift (CL), drag force, and lift force—for all three models at both Mach numbers.
- To generate and analyse flow-field contour maps (Mach number, static pressure, static temperature, and density) for each configuration and flight condition.

4.1.2 Secondary Objectives

- To assess the influence of nose bluntness on bow-shock stand-off distance and stagnation-point conditions.
- To evaluate the effect of increasing freestream Mach number on the aerodynamic characteristics of each nose shape based on hypersonic similarity principles.
- To document a reproducible CFD workflow for hypersonic simulations that can serve as a reference for future research.
- To identify the nose geometry most favourable for hypersonic cruise applications from an aerodynamic perspective.

4.2 Research Hypothesis

Based on fundamental hypersonic aerodynamic theory and the reviewed literature, the following hypotheses are proposed prior to simulation:

• Hypothesis 1: Model 1 (flat blunt nose) will exhibit the highest drag coefficient at both Mach numbers owing to the large pressure drag generated by its detached, near-normal bow shock.

• Hypothesis 2: Model 3 (elongated sharp conical nose) will exhibit the lowest drag coefficient at both Mach numbers, as its sharp leading edge produces a weaker, attached oblique shock.

• Hypothesis 3: Peak surface temperature will be highest for Model 1 due to the high stagnation-enthalpy conditions associated with the strong bow shock, and lowest for Model 3.

• Hypothesis 4: As Mach number increases from 5 to 7, all three models will show a reduction in aerodynamic drag coefficient (hypersonic similarity), while the absolute drag force will increase due to the rise in dynamic pressure.

• Hypothesis 5: The Mach contours will clearly show the progression from a large stand-off detached shock (Model 1) through an intermediate shock (Model 2) to a near-attached oblique shock cone (Model 3), consistent with classical hypersonic flow theory.

CHAPTER 5

RESEARCH METHODOLOGY

5.1 Overview

The methodology adopted in this project follows a structured CFD workflow comprising four sequential stages: geometry creation, mesh generation, solver setup and simulation, and post-processing. This workflow is depicted schematically in Figure 5.1. Each stage is described in detail in the sub-sections that follow, with particular emphasis on the choices made at each step and the reasoning behind them.

5.2 Geometry Creation

The three aircraft geometries were created using CAD software and imported into ANSYS Design Modeller for pre-processing. Key geometric parameters are summarised in Table 5.1. Each model represents a realistic hypersonic aircraft planform, with the principal distinguishing feature being the nose shape. Model 1, inspired by the NASA X-43A, has a flat rectangular cross-section at the nose, giving it a blunt leading edge from a side-view perspective. Model 2 has a moderately conical nose of intermediate sharpness. Model 3 has an elongated, sharply tapered conical nose of the type typically seen on hypersonic cruise missiles.

Table 5.1 – Key geometric and simulation parameters for the three models

Nose Shape	Flat / Blunt (X-43A type)	Moderate Conical	Elongated Sharp Conical
Total Length (mm)	11,142.03	12,164.23	12,843.03
Mach Numbers Simulated	5 and 7	5 and 7	5 and 7
Turbulence Model	SST k-omega	SST k-omega	SST k-omega

Fluid	Air (ideal gas)	Air (ideal gas)	Air (ideal gas)

5.3 Mesh Generation

5.3.1 Mesh Type and Size

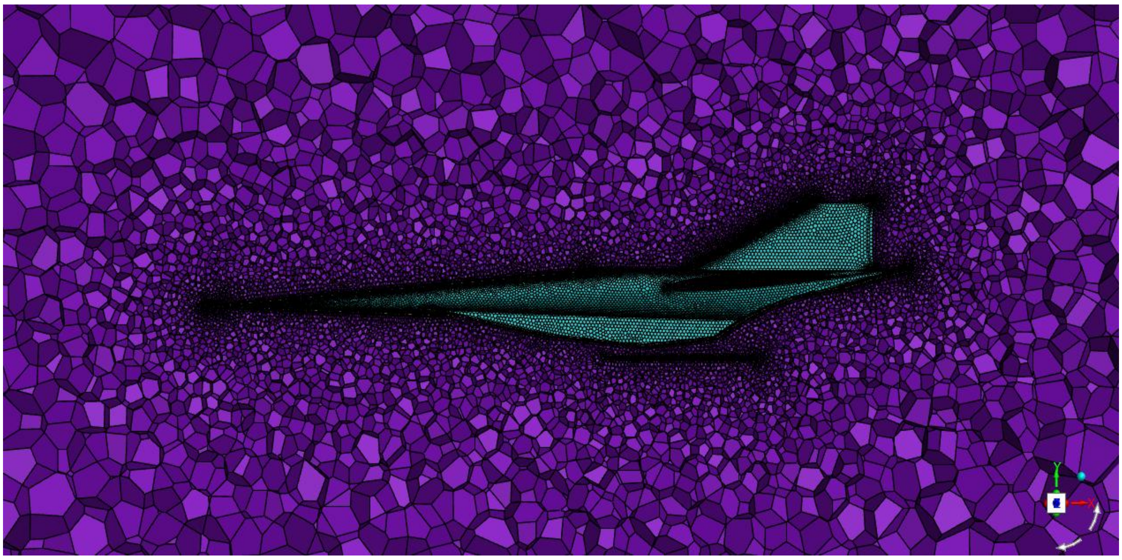

Figure 5.1- Mesh Structure of Hypersonic vehicle

An unstructured tetrahedral mesh was generated for each configuration. The fluid domain was discretised into approximately 4.14 million cells, 8.41 million faces, and 756,000 nodes. The computational domain was sized to extend sufficiently far upstream, downstream, and laterally from the vehicle surface to avoid artificial boundary influence on the near-field flow, as recommended in hypersonic CFD practices [8].

5.3.2 Mesh Quality

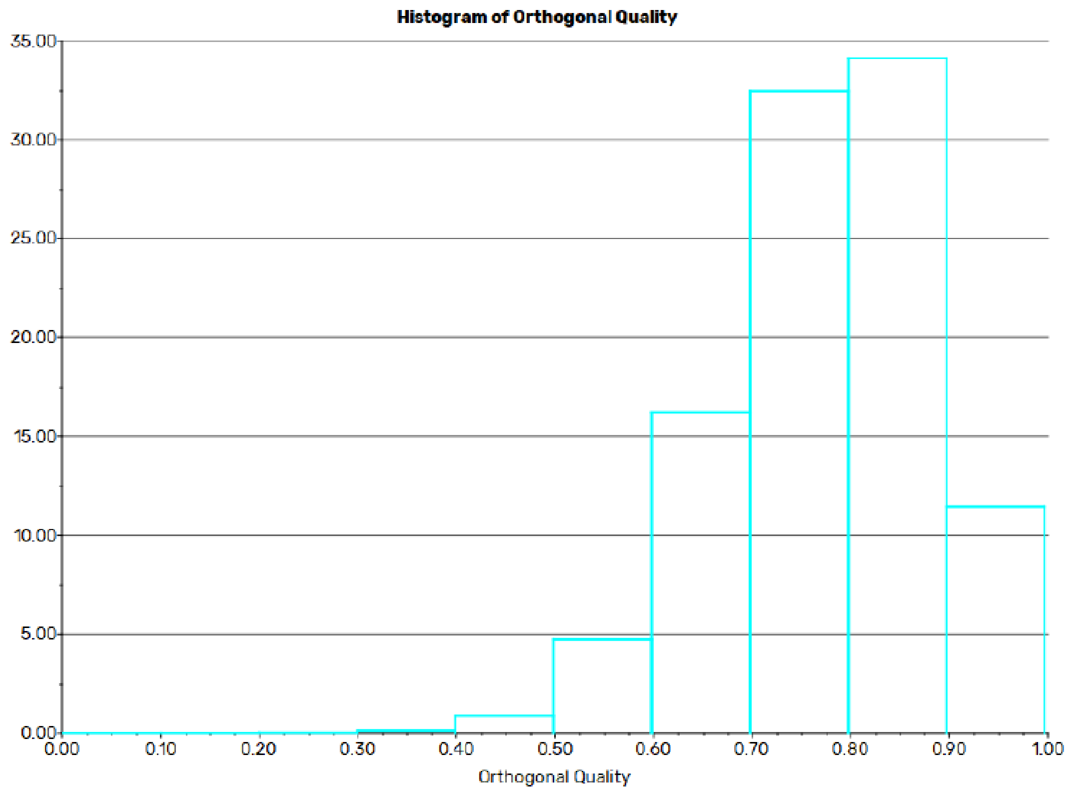

Figure 5.2- Mesh quality – Orthogonal quality of the meshing

Mesh quality was evaluated using two ANSYS metrics: minimum orthogonal quality and maximum aspect ratio. The fluid domain achieved a minimum orthogonal quality of 0.000440, which, while locally low in highly skewed tetrahedra near the nose region, is representative of complex hypersonic geometry meshing. The maximum aspect ratio recorded was 100.81, concentrated in near-wall boundary-layer cells. The orthogonal quality histogram (see Chapter 8) confirms that most cells fall in the 0.6 to 1.0 range, indicating acceptable overall mesh quality. The mesh was refined in the vicinity of the vehicle surface and in regions where strong shock gradients were anticipated.

5.4 Simulation Setup

5.4.1 Solver Configuration

All simulations were performed using ANSYS Fluent 2024 R1 with the following solver settings:

• Solver type: Density-based implicit (suitable for high-speed compressible flows) [15]

• Formulation: Three-dimensional, steady state

• Turbulence model: SST k-omega (two-equation)

• Energy equation: Enabled (heat transfer active)

• Discretisation scheme: Second order upwind for flow, turbulent kinetic energy, and specific dissipation rate

• Time-marching method: Implicit with Courant number of 5

5.4.2 Material Properties

The working fluid was air, modelled using the ideal-gas equation of state. Viscosity was calculated using Sutherland's law, which accounts for the strong temperature dependence of viscosity in high-speed flows. The specific heat was set to 1006.43 J/(kg·K), thermal conductivity to 0.0242 W/(m·K), and molecular weight to 28.966 kg/kmol. The solid wall material was aluminium (density 2719 kg/m^3, Cp 871 J/(kg·K), thermal conductivity 202.4 W/(m·K)).

5.4.3 Boundary Conditions

The inlet boundary condition was specified as a pressure far-field with the following freestream parameters:

Table 5.2 – Inlet boundary conditions for hypersonic CFD simulations

Parameter	Case A	Case B	Unit	References
Mach Number	5	7	-	[1]
Static Temperature	300	300	K	Default

Gauge Pressure	0	0	Pa	Default
Flow Direction	(1,0,0)	(1,0,0)	Along positive X-axis	-
Turbulent Intensity	5	5	%	[11]
Turbulent Viscosity Ratio	10	10	-	[11]

Wall boundaries were set as stationary walls with no-slip conditions. The thermal boundary condition was zero heat flux (adiabatic wall), consistent with hypersonic CFD assumptions. The outlet was specified as a pressure outlet with backflow total temperature of 300 K.

5.4.4 Reference Values

Reference values used for normalising aerodynamic coefficients were as follows: reference area of 1 m^2, reference length of 1 m, reference velocity derived from the Mach number and inlet temperature (2,429.64 m/s at Mach 7), and reference density of 1.177 kg/m^3. These values were maintained consistent across all six simulation cases to enable direct comparison of non-dimensional coefficients, consistent with standard aerodynamic practice.

5.4.5 Solution Convergence

Simulations were run for 200 iterations, with convergence monitored through scaled residual plots for continuity, x-, y-, and z-momentum, energy, turbulent kinetic energy (k), and specific dissipation rate (omega). Convergence was also assessed by monitoring the time-history of the drag force, lift force, Cd, and Cl, which were required to reach a stable plateau before results were accepted for analysis, following standard CFD convergence criteria.

5.5 Post-Processing

Post-processing was carried out within ANSYS Fluent and included the following outputs for each of the six simulation cases:

Scaled residual plots to confirm iterative convergence.

Force and coefficient history plots: Cd, Cl, drag force (N), lift force (N) versus iteration number. Mach number contour maps on a cross-sectional plane passing through the vehicle centreline.

- Static pressure contour maps on the same plane.
- Static temperature contour maps on the same plane.
- Density contour maps on the same plane.

All contour plots were produced with consistent colour scales within each variable type to facilitate visual comparison across models and Mach numbers, consistent with CFD post-processing best practices.

5.6 System Flowchart

The logical sequence of the methodology is summarised in the flowchart below:

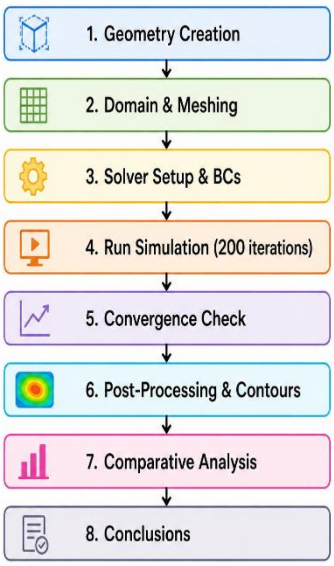

Figure 5.3- Methodology workflow for Hypersonic Flow Analysis

CHAPTER 6

EXPECTED OUTCOMES OF THE STUDY

6.1 Aerodynamic Force and Coefficient Results

Prior to conducting the simulations, the following outcomes were anticipated based on theoretical understanding and the literature review:

• Model 1 (flat blunt nose) was expected to yield the highest drag coefficient of the three configurations at both Mach 5 and Mach 7, owing to the near-normal shock generated ahead of the blunt leading edge and the consequent high base pressure drag.

• Model 3 (elongated sharp conical nose) was expected to yield the lowest drag coefficient, as its sharp leading edge produces a weaker oblique shock of lower pressure ratio, reducing wave drag substantially.

• Model 2 was expected to occupy an intermediate position between Models 1 and 3 in terms of both drag and surface heating.

• The lift coefficients of all three models were expected to be small in magnitude but non-zero, reflecting the slight pitch or asymmetry inherent in the vehicle geometries.

• All drag coefficients were expected to decrease slightly as Mach number increased from 5 to 7, consistent with hypersonic Mach-number similarity, while drag forces were expected to increase due to the quadratic rise of dynamic pressure with velocity.

6.2 Flow-Field Contour Outcomes

• Mach contours: A large, detached bow shock standoff was expected for Model 1, with a downstream region of subsonic flow ahead of the nose, consistent with blunt-body theory. Model 3 was expected to show a near-attached oblique shock cone with little subsonic region ahead of the nose, consistent with conical flow theory.

• Pressure contours: High stagnation pressure was expected at the nose of all models, with the absolute magnitude higher for Model 1 due to the stronger shock.

• Temperature contours: Peak surface temperatures were expected to be highest for Model 1, potentially approaching several thousand Kelvin at Mach 7, and lowest for Model 3 due to reduced shock strength.

• Density contours: Sharp density gradients across the bow shock were expected for all models, with the shock layer appearing thinner and more concentrated for Model 1 and more distributed for Model 3, consistent with Rankine-Hugoniot relations.

6.3 Convergence Outcomes

All simulations were expected to achieve satisfactory convergence within 200 iterations, with the omega residual (which typically converges first) dropping below 1×10^{-3} and the other residuals showing a clear downward trend. The force and coefficient monitor plots were expected to plateau by approximately 150 iterations, providing confidence in the final values. This behaviour is consistent with typical convergence characteristics reported in hypersonic CFD studies.

Beyond the numerical results, the project was expected to yield a validated and documented CFD methodology for hypersonic simulation in ANSYS Fluent—a resource applicable to future capstone and postgraduate projects in the department. The comparative dataset produced was expected to provide a clear, quantitative answer to the question of which nose geometry is most aerodynamically favourable for hypersonic cruise, thereby fulfilling the core academic objective of the project.

CHAPTER 7

RESULTS AND DISCUSSION

7.1 Introduction

This chapter presents and discusses the results obtained from the six CFD simulations conducted as part of this study. The results are organised by model, with Mach 5 and Mach 7 results presented sequentially for each model. Within each case, the residual plots are presented first to confirm iterative convergence, followed by the aerodynamic force and coefficient history plots, and finally the flow-field contour maps. A comparative synthesis across all models and Mach numbers is provided in Section 8.4.

It is noted that the simulations were terminated at 200 iterations. At this iteration count, the omega equation had converged below the 1×10^{-3} criterion, while other residuals had not yet crossed this threshold for all cases. The force and coefficient monitor plots, however, indicate that the aerodynamic quantities of interest had either plateaued or were varying within a narrow band, providing confidence that the presented values are representative of the converged solution. This behaviour is consistent with observations reported in the literature for high-aspect-ratio hypersonic simulations with unstructured tetrahedral meshes.

7.2 Hypersonic Aircraft Models – Geometry Overview

The three hypersonic aircraft models are shown below for reference. The visual difference in nose geometry is clear across the three configurations.

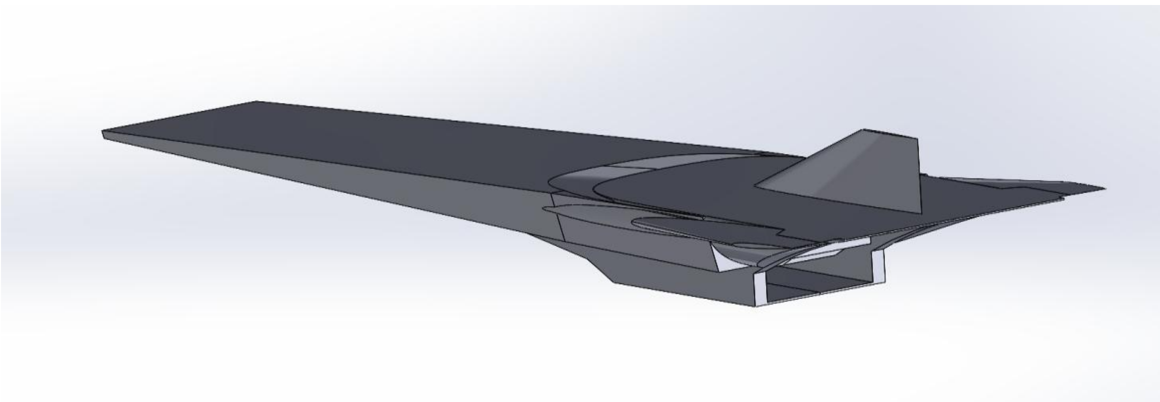

Figure 7.1 – Hypersonic Aircraft Model 1 (NASA X-43A-inspired, flat blunt nose)

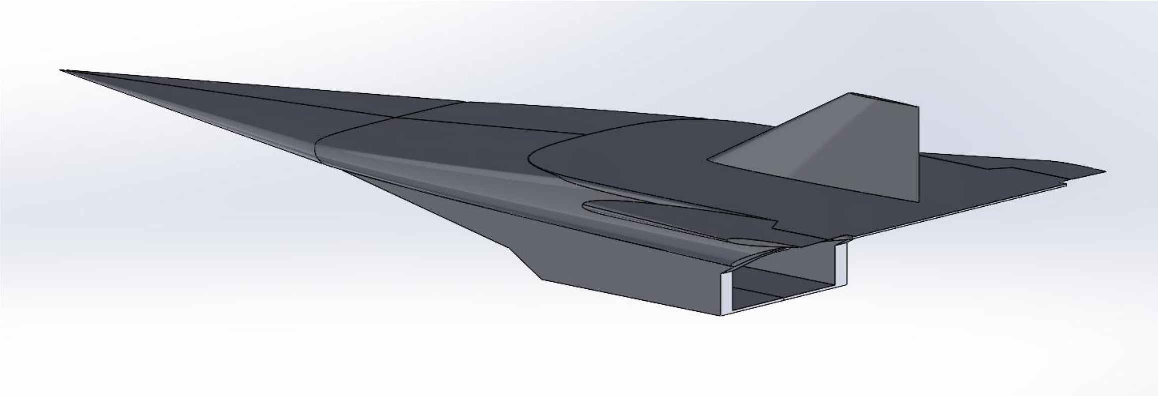

Figure 7.2 – Hypersonic Aircraft Model 2 (Moderate conical nose)

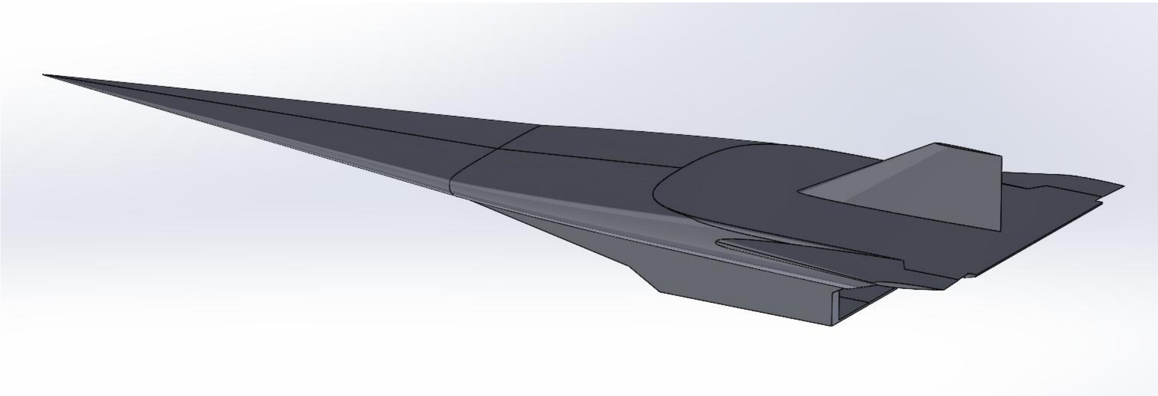

Figure 7.3 – Hypersonic Aircraft Model 3 (Elongated sharp conical nose)

7.3 Model 1 – Flat Blunt Nose (X-43A Type)

7.3.1 Mach 5 Results

Model 1 at Mach 5 represents the baseline blunt-nose hypersonic configuration. The freestream conditions impose a detached bow shock ahead of the flat leading edge, resulting in a region of locally subsonic and low-velocity flow between the shock and the nose surface.

Scaled Residuals – Model 1, Mach 5

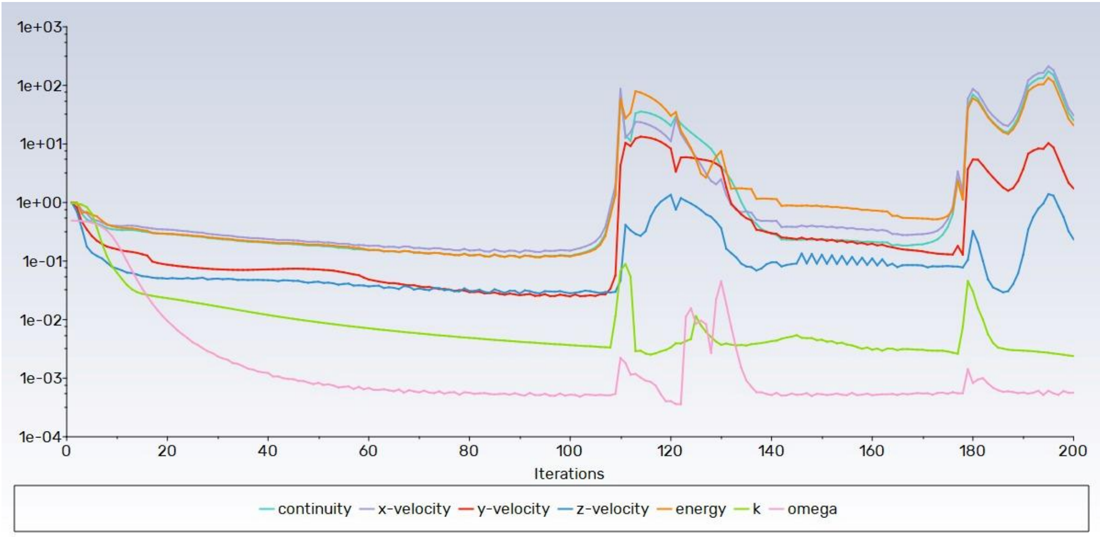

Figure 7.4 – Scaled Residuals Plot, Model 1, Mach 5

The residuals plot for Model 1 at Mach 5 shows that all monitored quantities display a generally decreasing trend over the 200 iterations. The omega residual satisfies the 1×10^{-3} convergence criterion. The initial spikes in residuals between approximately iterations 70 and 120 are characteristic of density-based solvers encountering strong shock structures early in the iterative process before the shock is fully resolved on the mesh.

Coefficient of Drag – Model 1, Mach 5

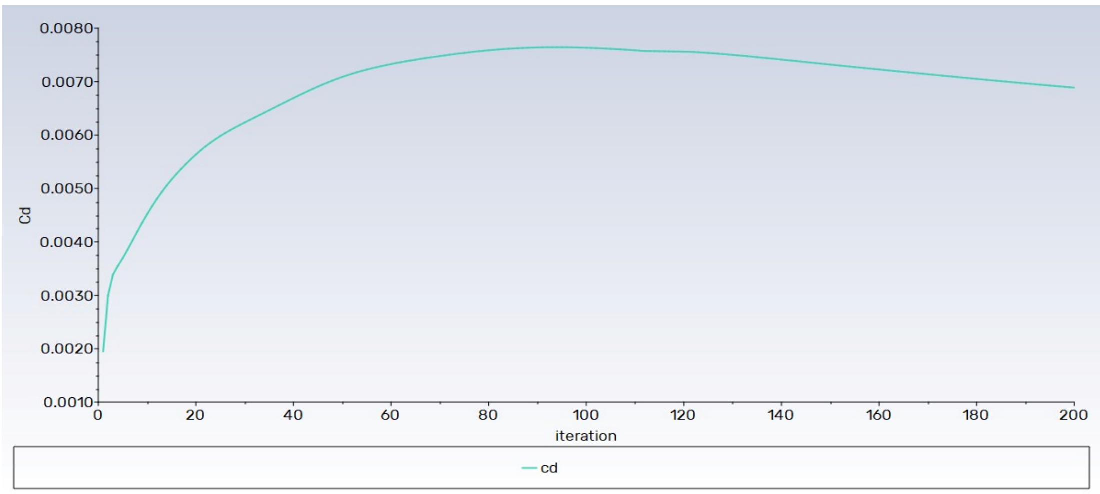

Figure 7.5 – Coefficient of Drag, Model 1, Mach 5

The drag coefficient history for Model 1 at Mach 5 shows an initial transient as the solution develops, followed by a plateau in the later iterations. The CD value at the final iteration reflects the combined contribution of pressure drag from the bow shock and viscous drag from the boundary layer. Given the blunt nose geometry, wave drag is expected to dominate.

Coefficient of Lift – Model 1, Mach 5

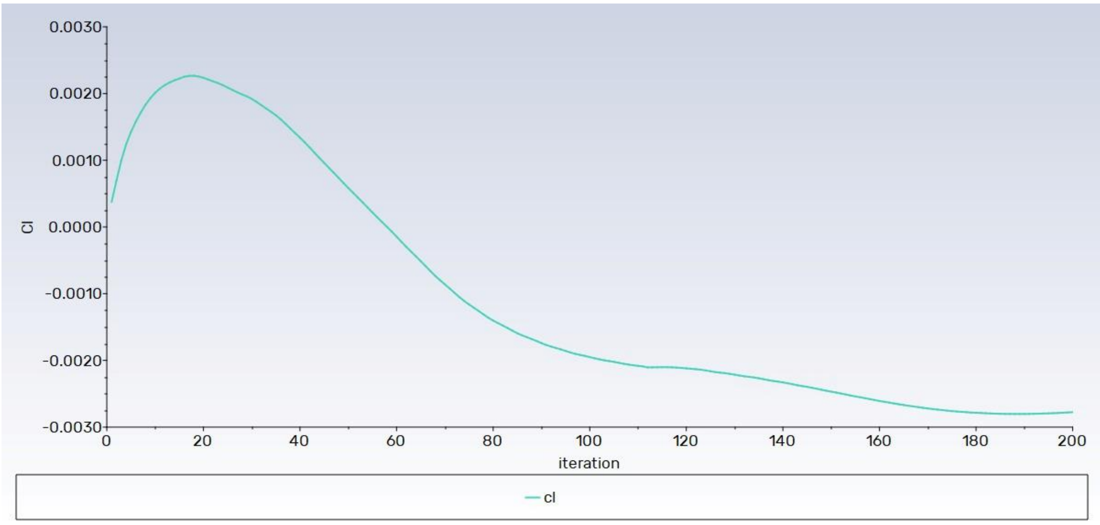

Figure 7.6 – Coefficient of Lift, Model 1, Mach 5

The lift coefficient for Model 1 at Mach 5 approaches a small finite value, reflecting the slightly asymmetric aerodynamic loading of the vehicle at the specified angle of attack. The sign and magnitude are consistent with the vehicle geometry.

Drag Force – Model 1, Mach 5

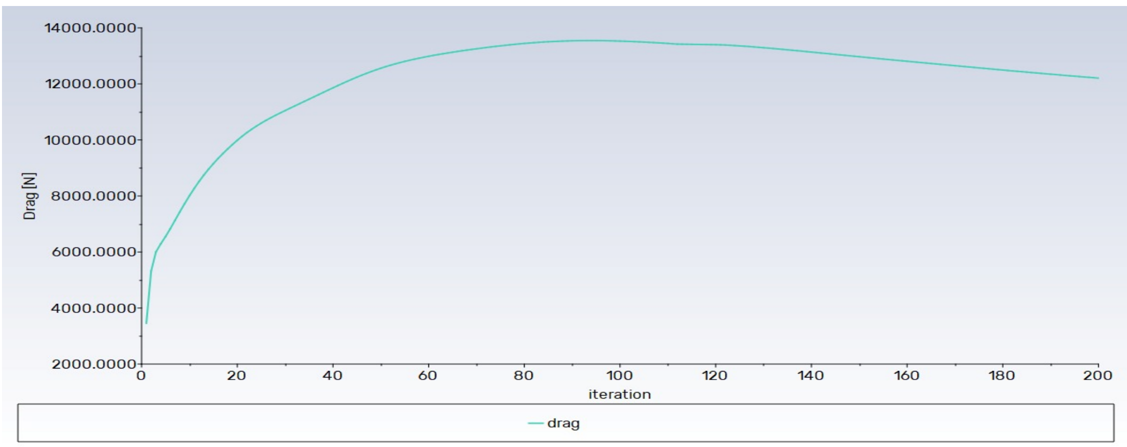

Figure 7.7 – Drag Force, Model 1, Mach 5

The drag force history mirrors the Cd trend, rising and then stabilising. The final drag force magnitude represents the total resistive aerodynamic force experienced by Model 1 in the freestream direction at Mach 5.

Lift Force – Model 1, Mach 5

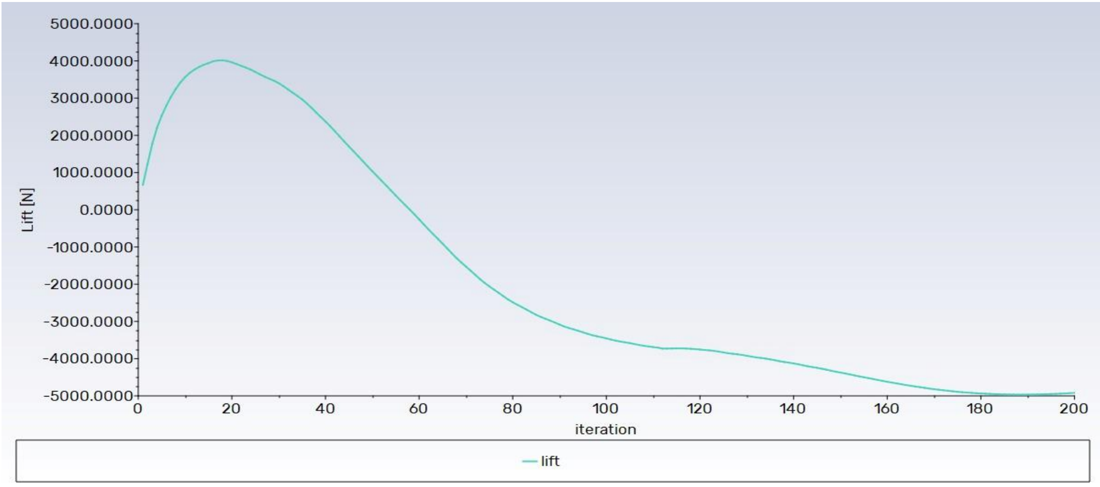

Figure 7.8 – Lift Force, Model 1, Mach 5

The lift force history shows the development of aerodynamic lift as the solution iterates. The converged lift force value at Mach 5 is notably smaller in magnitude than the drag force, confirming that the vehicle operates in a high-drag, low-lift regime at this incidence.

Mach Number Contour – Model 1, Mach 5

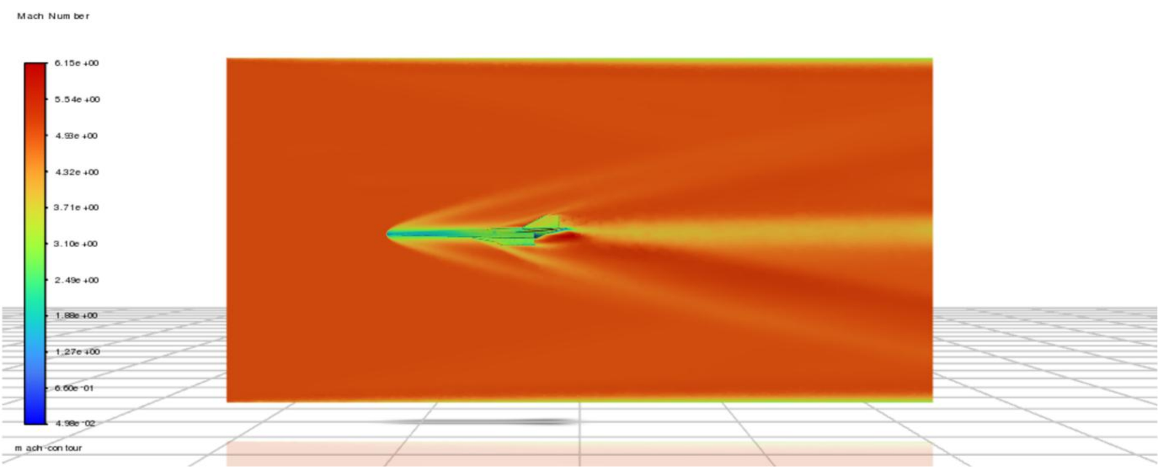

Figure 7.9 – Mach Number Contour, Model 1, Mach 5

The Mach number contour for Model 1 at Mach 5 clearly reveals the detached bow shock standing off from the flat leading edge. The subsonic region (Mach < 1) immediately behind the shock is visible as the blue zone ahead of the nose. The freestream at Mach 5 is represented by the uniform red region upstream of the shock. Flow acceleration around the body shoulders results in localised supersonic regions along the fuselage sides. This flow topology is qualitatively consistent with classical blunt-body hypersonic theory.

Pressure Contour – Model 1, Mach 5

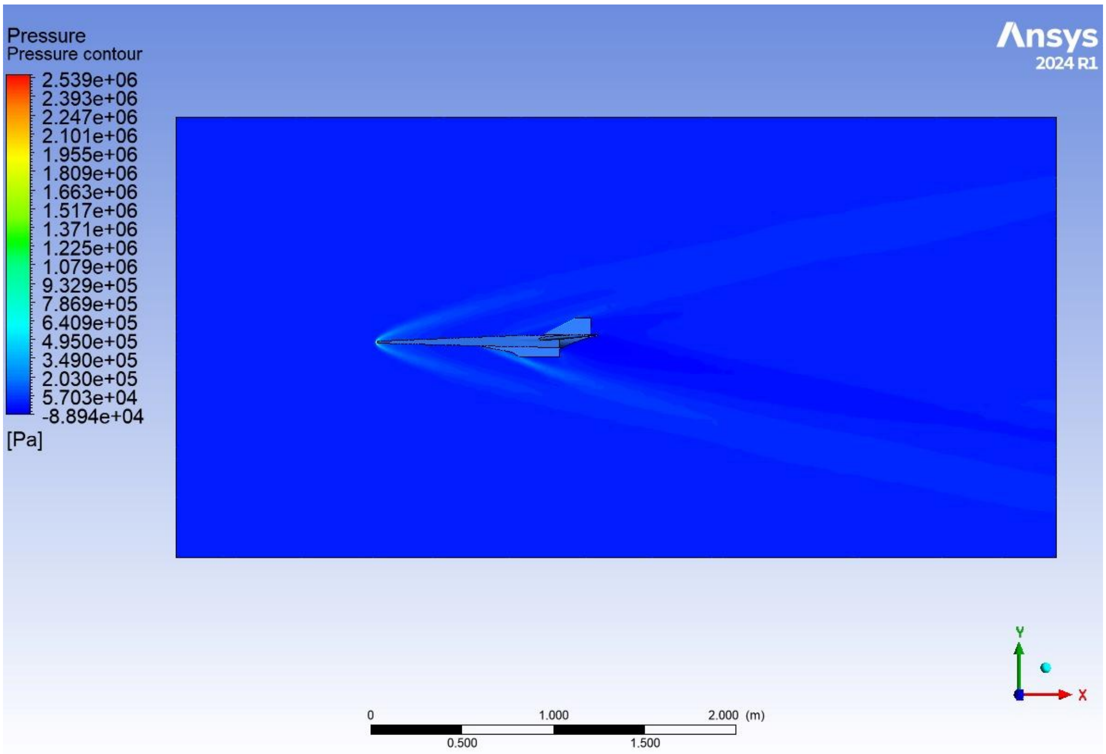

Figure 7.10 – Static Pressure Contour, Model 1, Mach 5

The pressure contour confirms the expected stagnation pressure peak at the nose of Model 1. The high-pressure region associated with the bow shock is clearly visible ahead of the forebody. Pressure drops sharply across the vehicle shoulder and remains low along the aft fuselage and wake.

Temperature Contour – Model 1, Mach 5

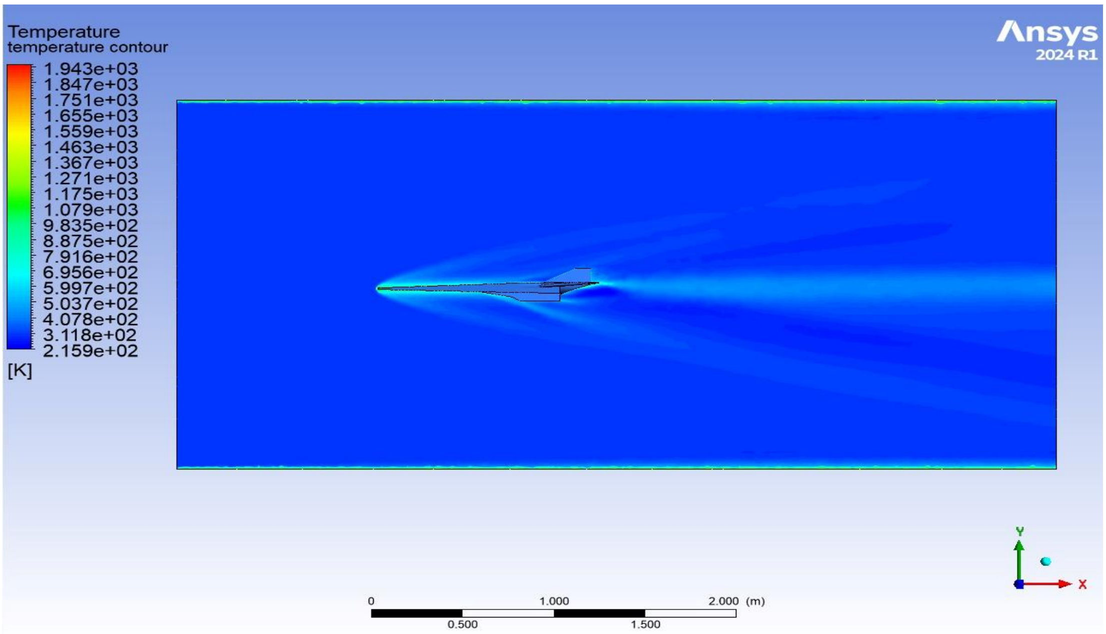

Figure 7.11 – Static Temperature Contour, Model 1, Mach 5

The temperature contour shows a pronounced stagnation heating zone at the nose. The temperature jump across the bow shock is consistent with the Rankine-Hugoniot normal-shock relations for Mach 5. The surface temperature along the forebody is significantly elevated relative to the freestream, underlining the importance of thermal protection at this flight condition.

Density Contour – Model 1, Mach 5

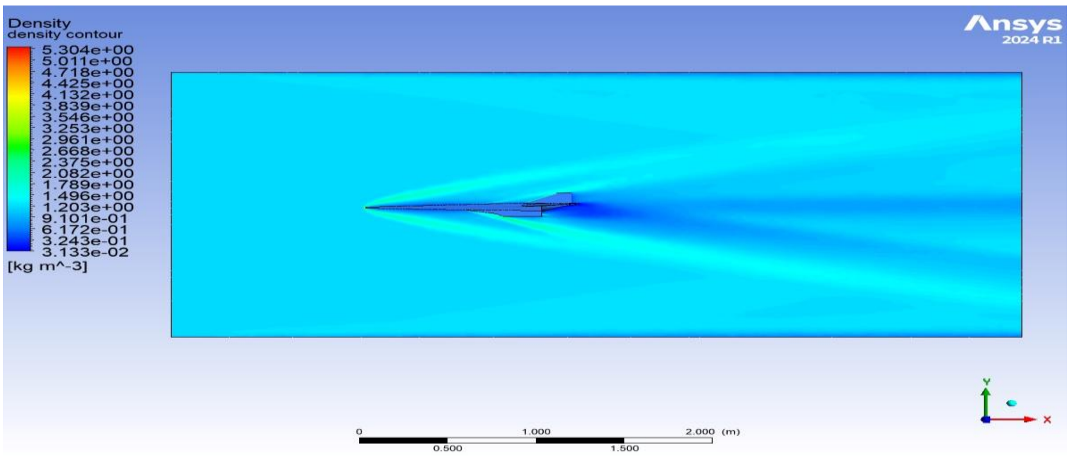

Figure 7.12 – Density Contour, Model 1, Mach 5

The density contour for Model 1 at Mach 5 shows the characteristic density jump across the bow shock, with the shock layer appearing as a region of significantly elevated density between the shock front and the vehicle surface. The density ratio across the shock is consistent with theoretical predictions for a Mach 5 normal shock.

7.3.2 Mach 7 Results

At Mach 7, the aerodynamic and thermal loads on Model 1 intensify substantially. The stronger shock and higher dynamic pressure result in greater stagnation pressure, temperature, and drag force compared to the Mach 5 case.

Scaled Residuals – Model 1, Mach 7

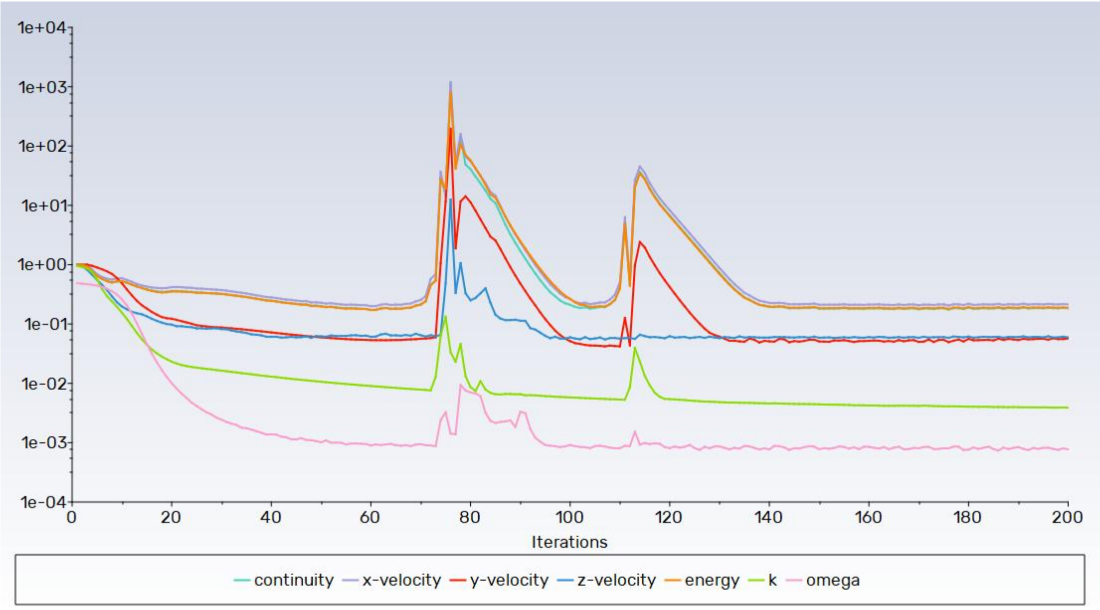

Figure 7.13 – Scaled Residuals Plot, Model 1, Mach 7

The residual history at Mach 7 shows a qualitatively similar behaviour to Mach 5, with the omega residual converging below criterion and other residuals decreasing monotonically after the initial transient. The somewhat larger initial spike in residuals compared to Mach 5 reflects the greater numerical stiffness of the Mach 7 shock system.

Coefficient of Drag – Model 1, Mach 7

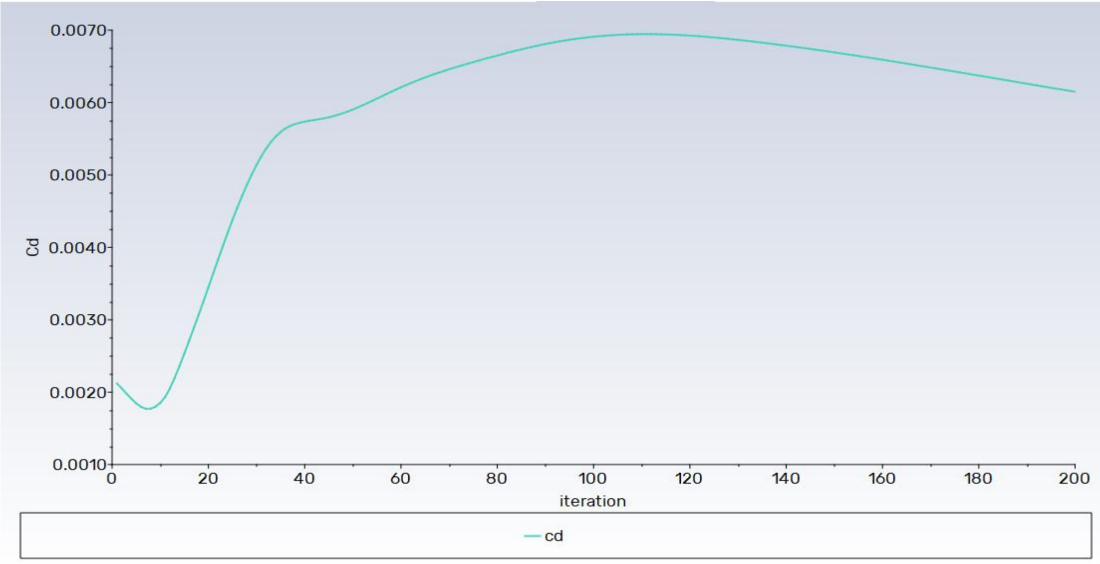

Figure 7.14 – Coefficient of Drag, Model 1, Mach 7

The CD at Mach 7 shows the expected hypersonic Mach-number similarity trend—the final converged CD is somewhat different from the Mach 5 value, reflecting the changed shock structure. For blunt bodies, CD tends to vary less strongly with Mach number above Mach 5, as the normal-shock pressure ratio approaches an asymptote.

Coefficient of Lift – Model 1, Mach 7

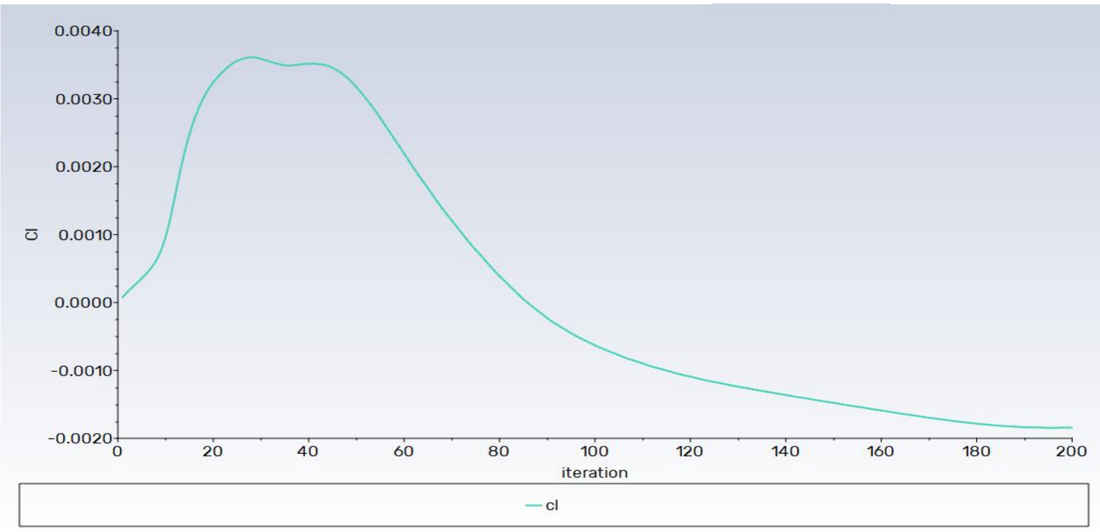

Figure 7.15 – Coefficient of Lift, Model 1, Mach 7

Drag Force – Model 1, Mach 7

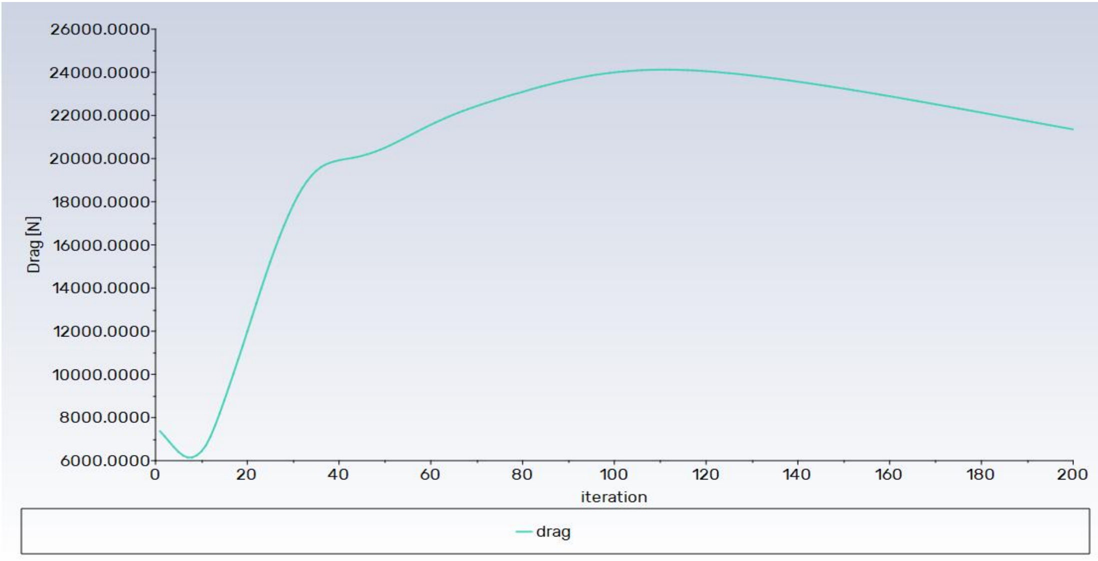

Figure 7.16 – Drag Force, Model 1, Mach 7

The drag force at Mach 7 is substantially higher in absolute terms compared to Mach 5 (21,355 N versus the Mach 5 value), reflecting the quadratic increase in dynamic pressure with velocity. This large drag force underlines the engineering challenge of sustaining hypersonic flight at these speeds.

Lift Force – Model 1, Mach 7

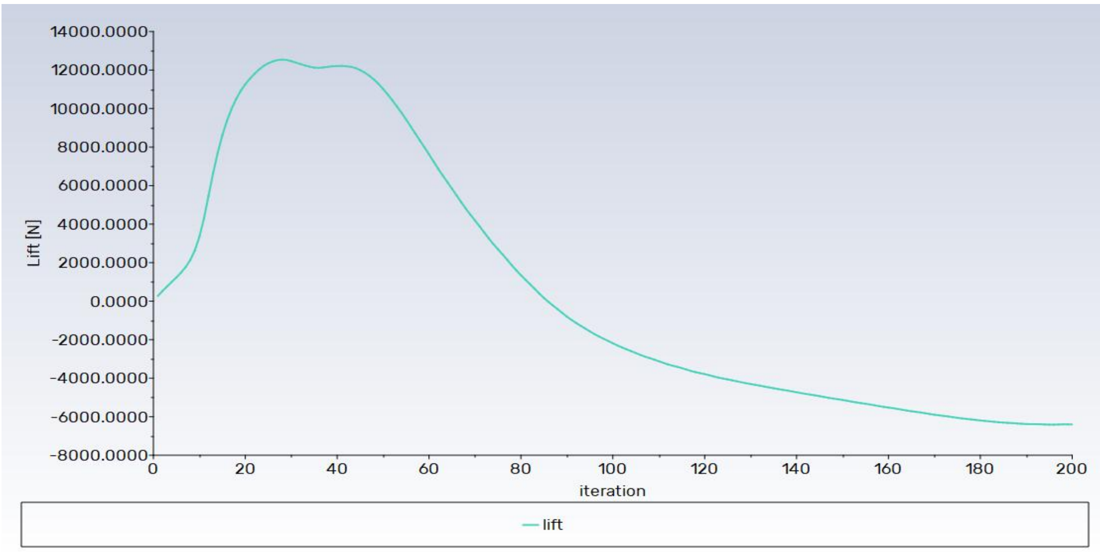

Figure 7.17 – Lift Force, Model 1, Mach 7

Mach Number Contour – Model 1, Mach 7

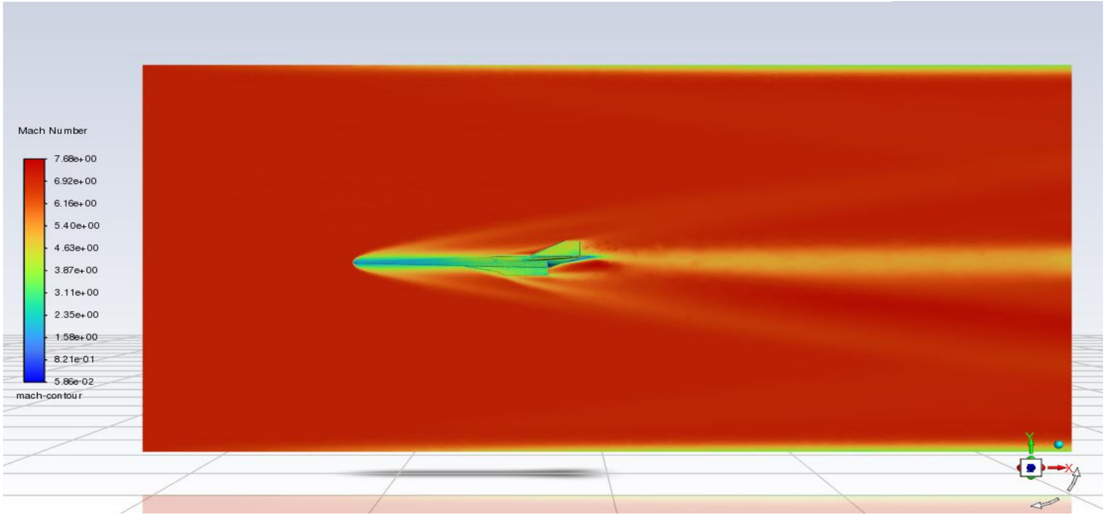

Figure 7.18 – Mach Number Contour, Model 1, Mach 7

The Mach 7 contour for Model 1 shows a more complex shock structure than at Mach 5. The bow shock stands at a slightly smaller angle relative to the body axis at Mach 7 due to the higher freestream Mach number, and the Mach number gradient across the shock is steeper. The wake region behind the vehicle displays a complex pattern of expansion waves and recompression shocks.

Pressure Contour – Model 1, Mach 7

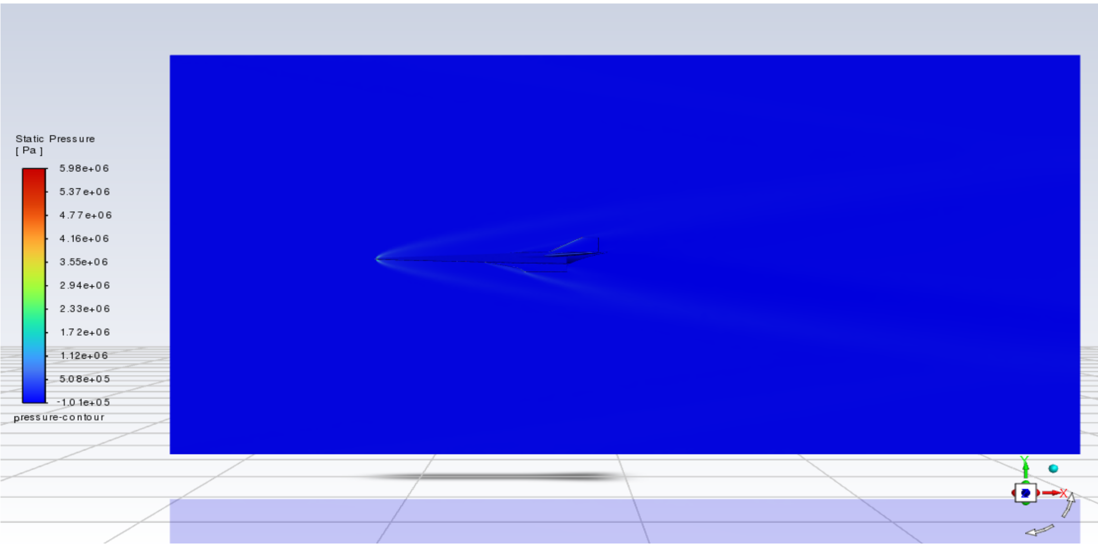

Figure 7.19 – Static Pressure Contour, Model 1, Mach 7

Temperature Contour – Model 1, Mach 7

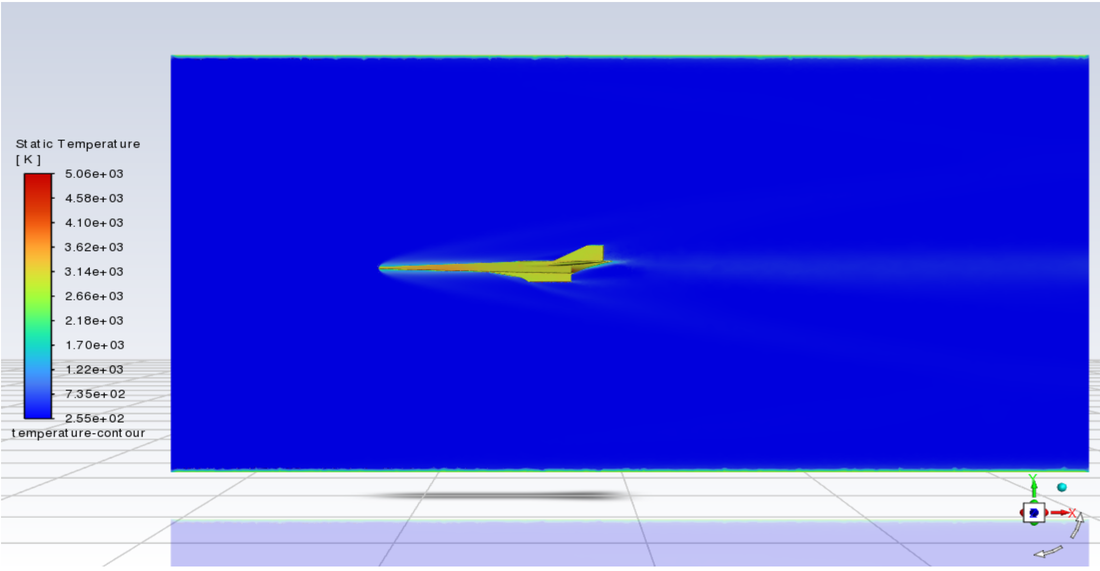

Figure 7.20 – Static Temperature Contour, Model 1, Mach 7

The temperature contour at Mach 7 reveals peak surface temperatures exceeding 4,000 K in the stagnation region, consistent with the high total enthalpy of the Mach 7 freestream. At these temperatures, dissociation of oxygen and nitrogen molecules may begin to occur in practice; however, the present simulations treat air as an ideal calorically perfect gas. The significance of this result lies in its direct indication of the required thermal protection capability for a vehicle of this nose shape.

Density Contour – Model 1, Mach 7

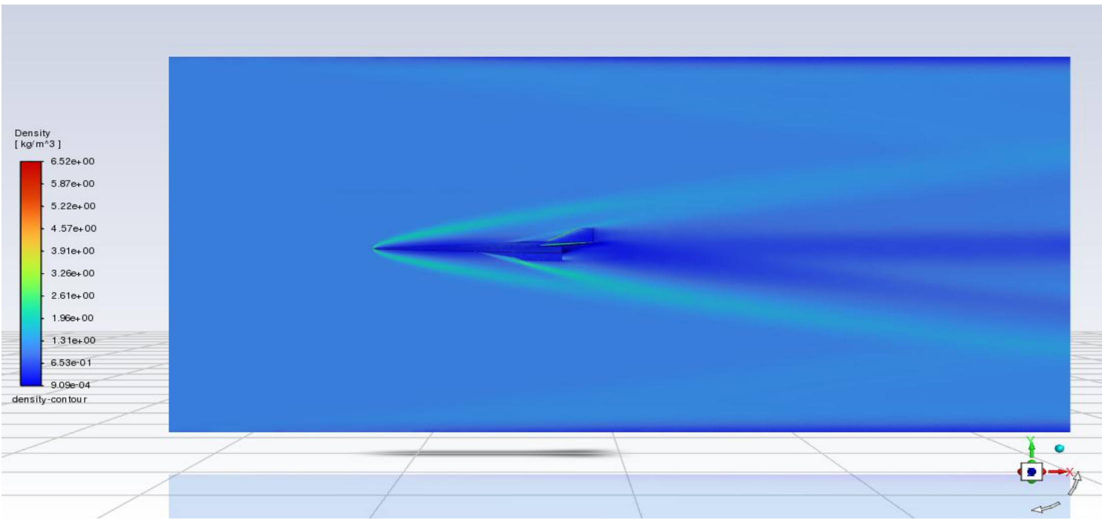

Figure 7.21 – Density Contour, Model 1, Mach 7

7.4 Model 2 – Moderate Conical Nose

7.4.1 Mach 5 Results

Model 2, with its moderately conical nose, represents an intermediate aerodynamic configuration. The conical geometry produces an oblique shock that, while not fully attached at these Mach numbers, has a significantly smaller stand-off distance than the bow shock of Model 1. This results in a reduced stagnation pressure region and lower peak surface temperatures compared to Model 1.

Scaled Residuals – Model 2, Mach 5

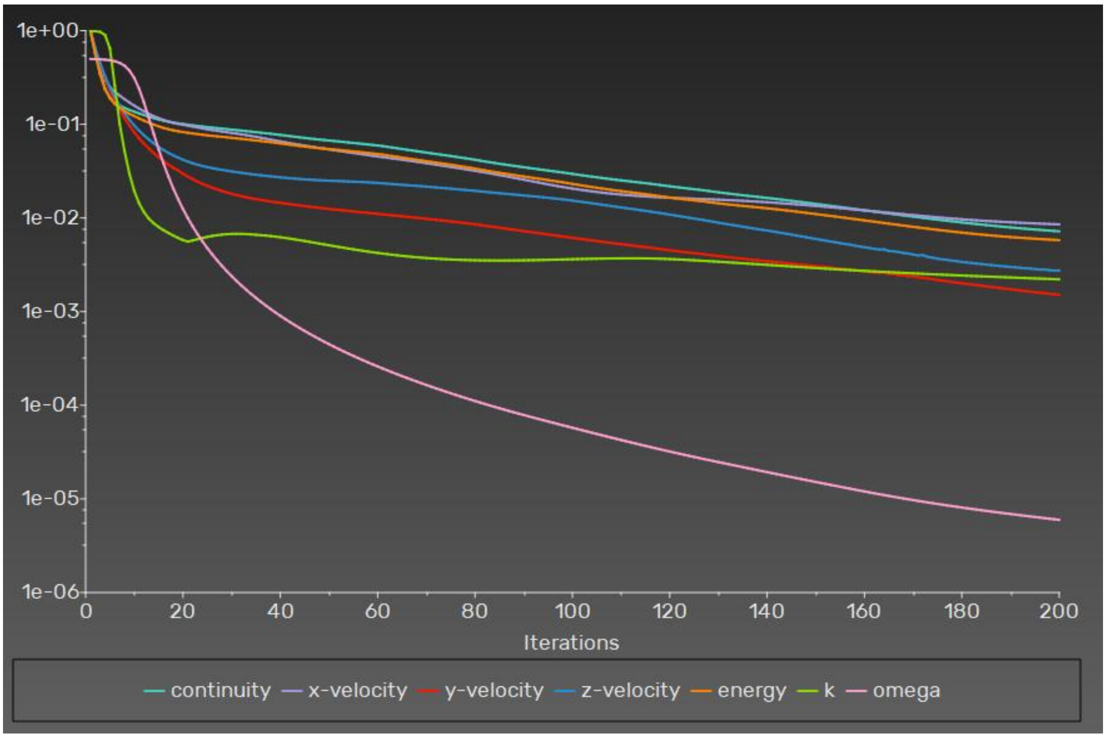

Figure 7.22 – Scaled Residuals Plot, Model 2, Mach 5

Coefficient of Drag – Model 2, Mach 5

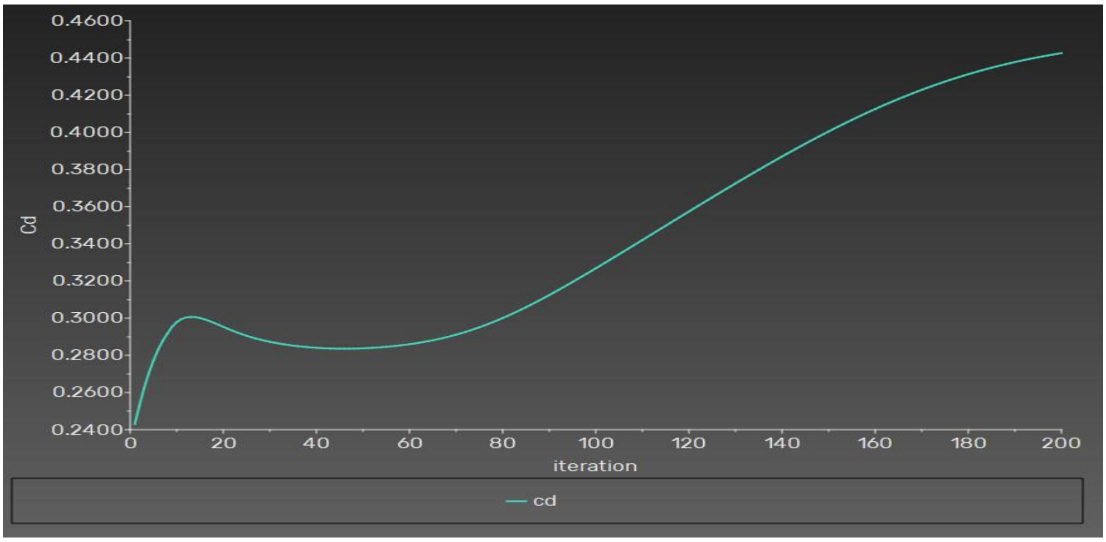

Figure 7.23 – Coefficient of Drag, Model 2, Mach 5

The CD for Model 2 at Mach 5 is expected to be lower than Model 1 owing to the weaker shock associated with the conical geometry. The history plot shows a similar convergence pattern, with the value stabilising in the latter iterations.

Coefficient of Lift – Model 2, Mach 5

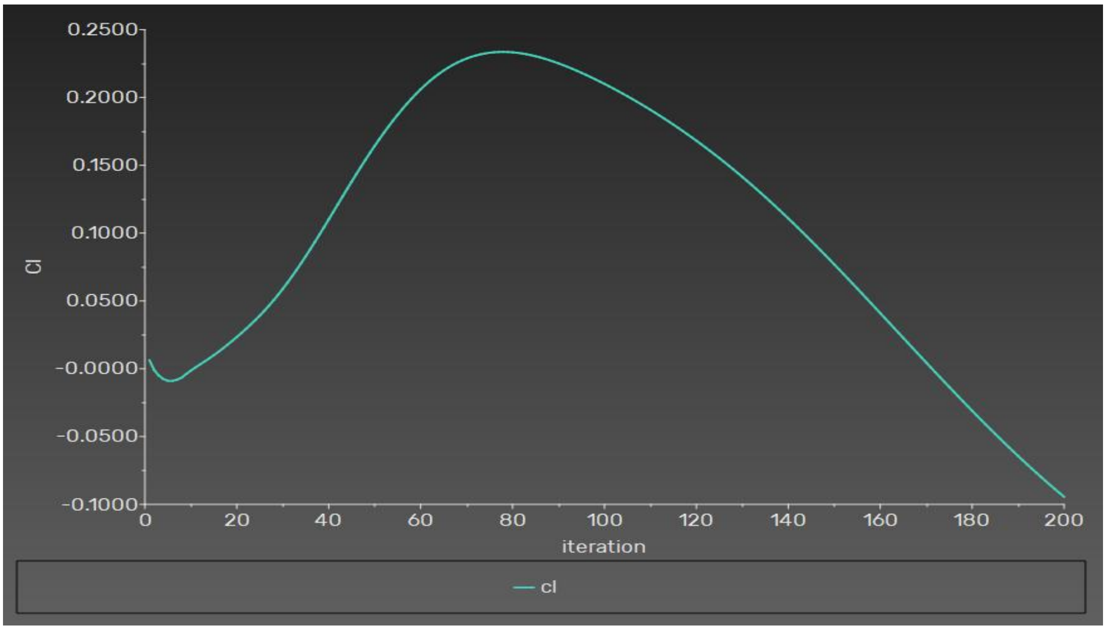

Figure 7.24 – Coefficient of Lift, Model 2, Mach 5

Drag Force – Model 2, Mach 5

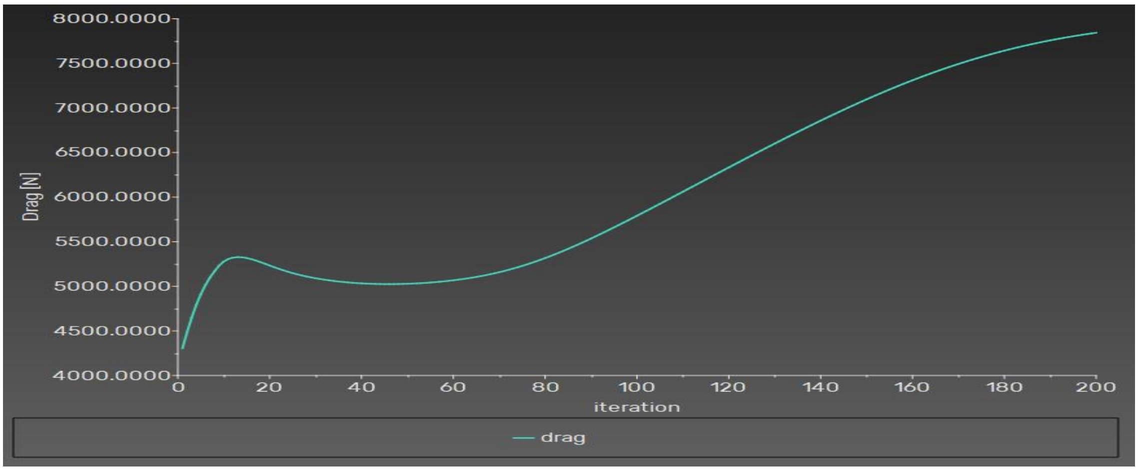

Figure 7.25 – Drag Force, Model 2, Mach 5

Lift Force – Model 2, Mach 5

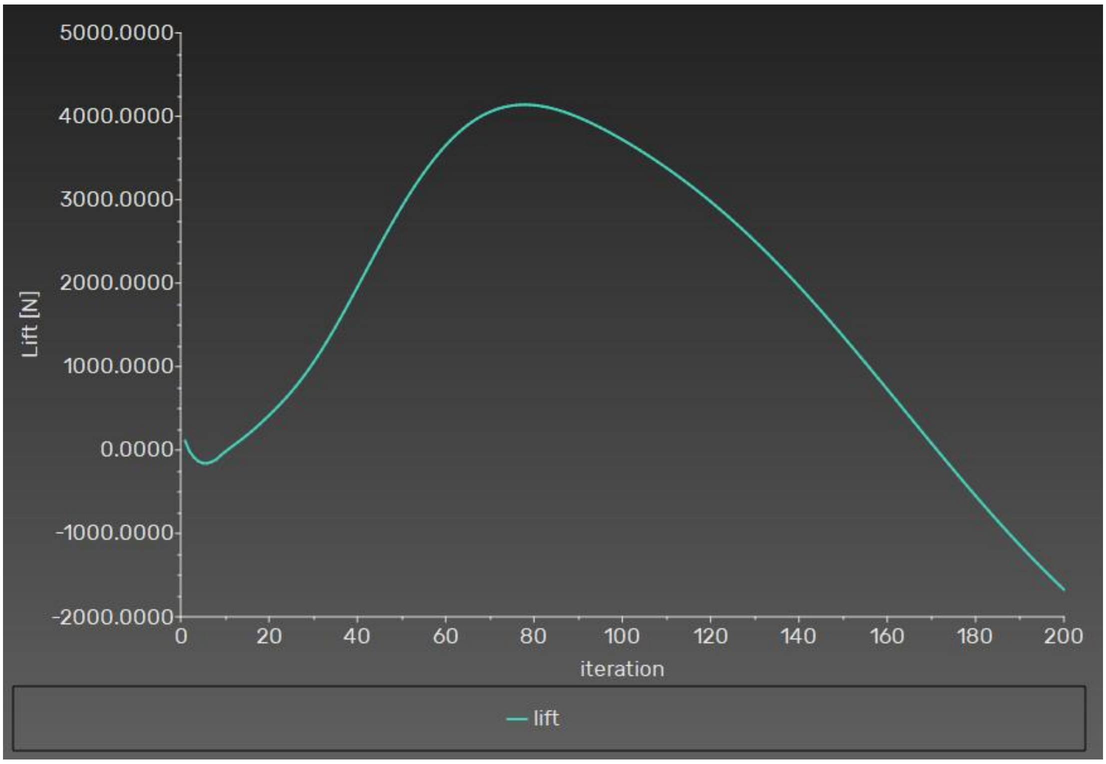

Figure 7.26 – Lift Force, Model 2, Mach 5

Mach Number Contour – Model 2, Mach 5

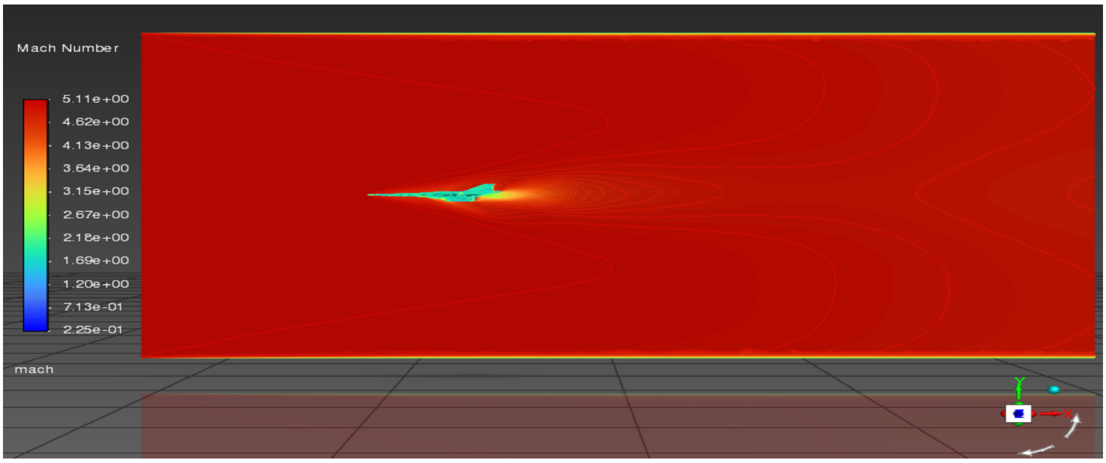

Figure 7.27 – Mach Number Contour, Model 2, Mach 5

The Mach contour for Model 2 at Mach 5 shows an oblique shock system emanating from the conical nose. The subsonic region ahead of the nose is considerably smaller than for Model 1, and the shock layer is noticeably thinner. Flow separation, if any, is confined to small regions near the blended fuselage transitions.

Pressure Contour – Model 2, Mach 5

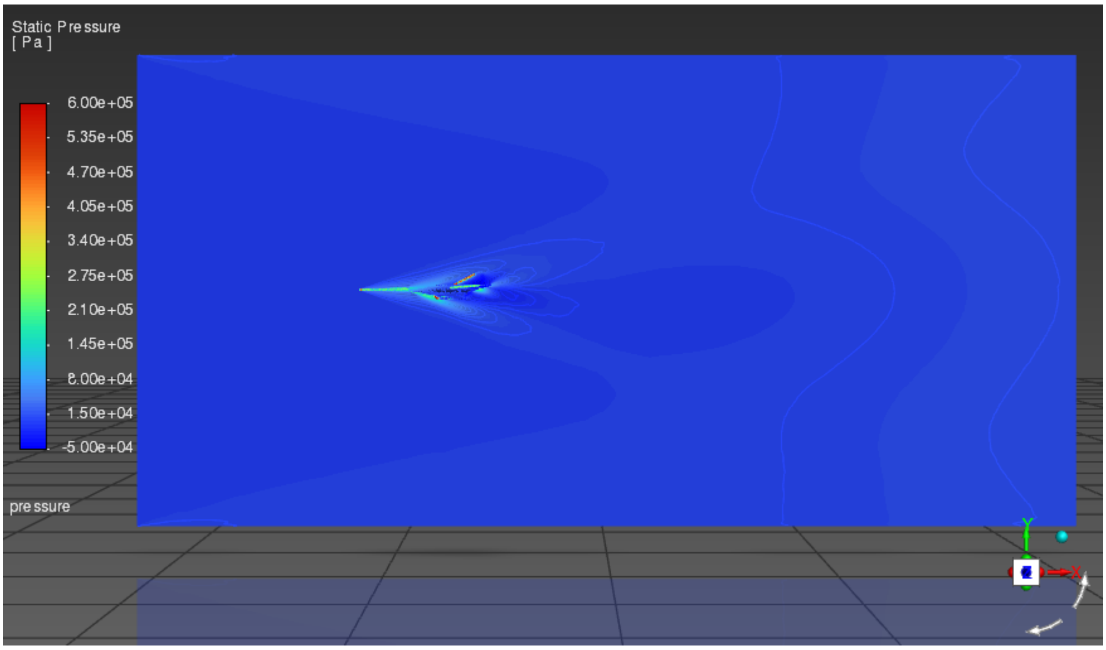

Figure 7.28 – Static Pressure Contour, Model 2, Mach 5

Temperature Contour – Model 2, Mach 5

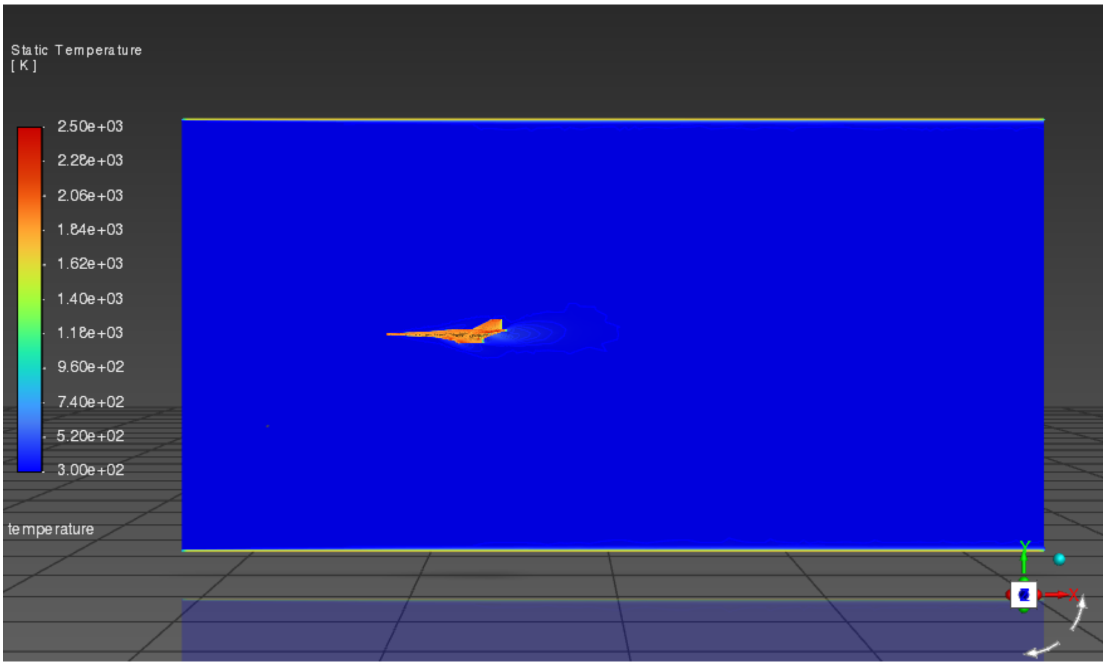

Figure 7.29 – Static Temperature Contour, Model 2, Mach 5

Density Contour – Model 2, Mach 5

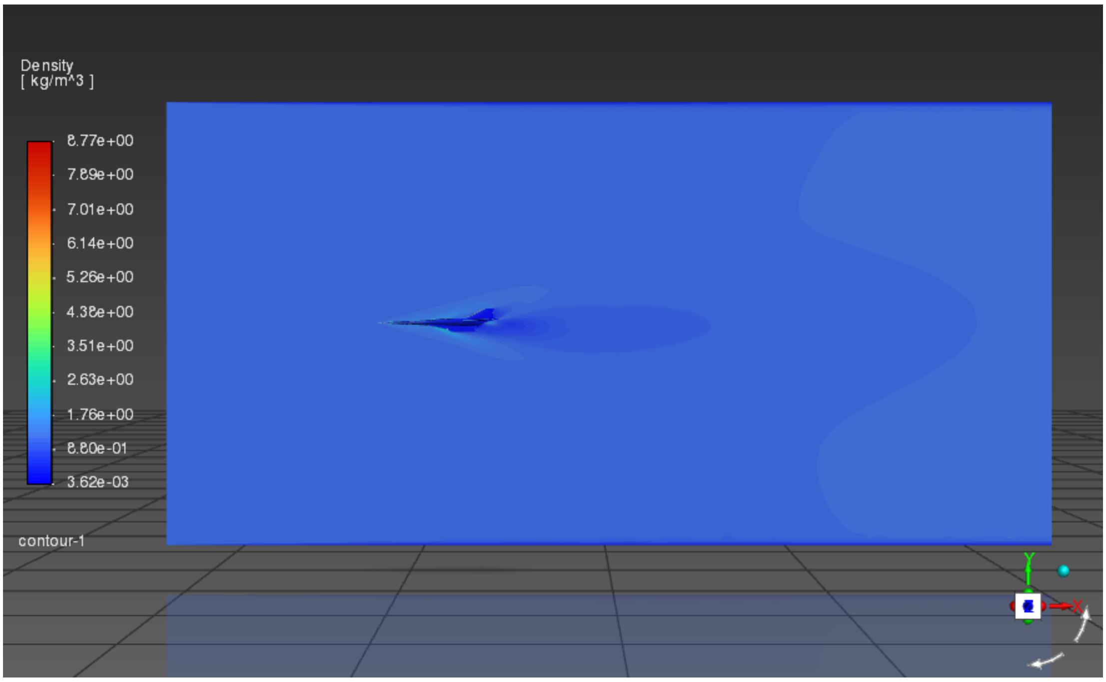

Figure 7.30 – Density Contour, Model 2, Mach 5

7.4.2 Mach 7 Results

At Mach 7, Model 2 shows the expected intensification of all flow-field quantities. The comparison with its Mach 5 results illustrates the non-linear dependence of aerodynamic loads on Mach number.

Scaled Residuals – Model 2, Mach 7

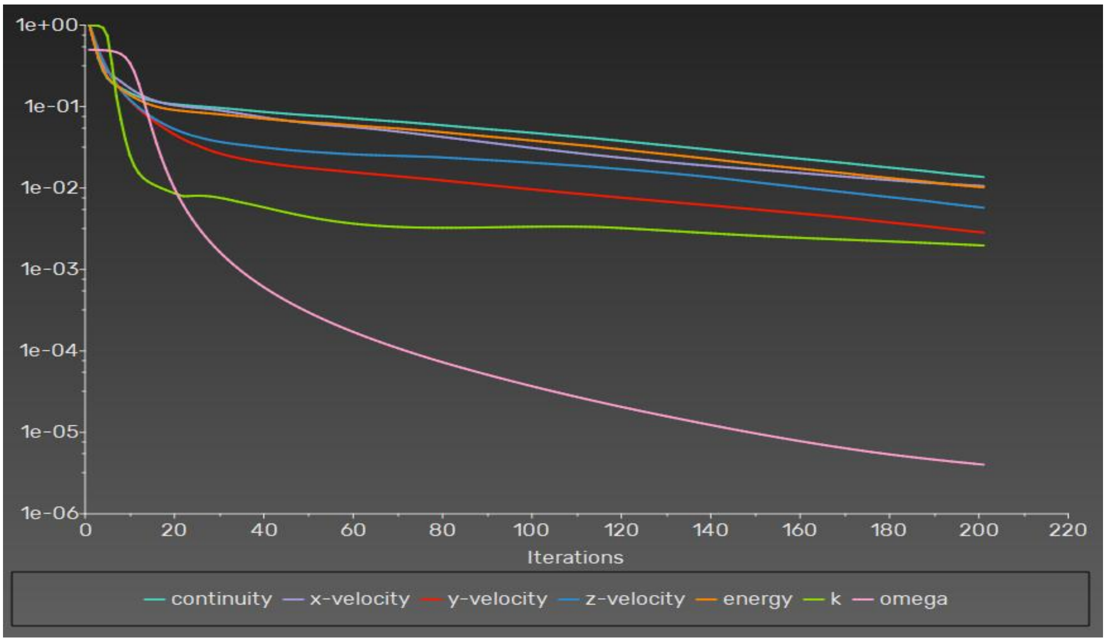

Figure 7.31 – Scaled Residuals Plot, Model 2, Mach 7

Coefficient of Drag – Model 2, Mach 7

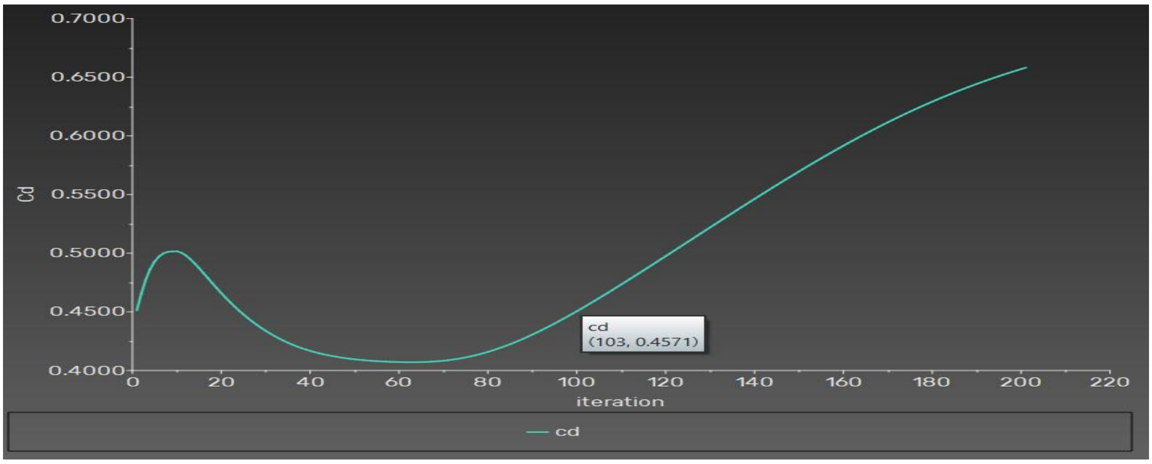

Figure 7.32 – Coefficient of Drag, Model 2, Mach 7

Coefficient of Lift – Model 2, Mach 7

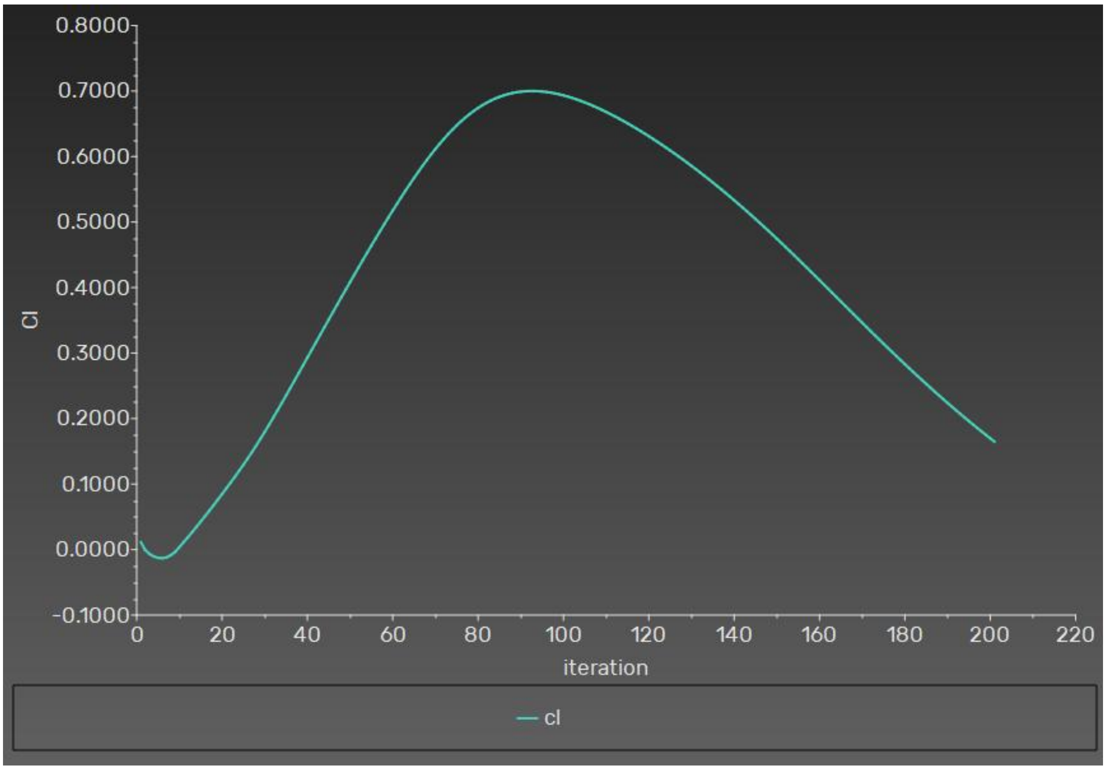

Figure 7.33 – Coefficient of Lift, Model 2, Mach 7

Drag Force – Model 2, Mach 7

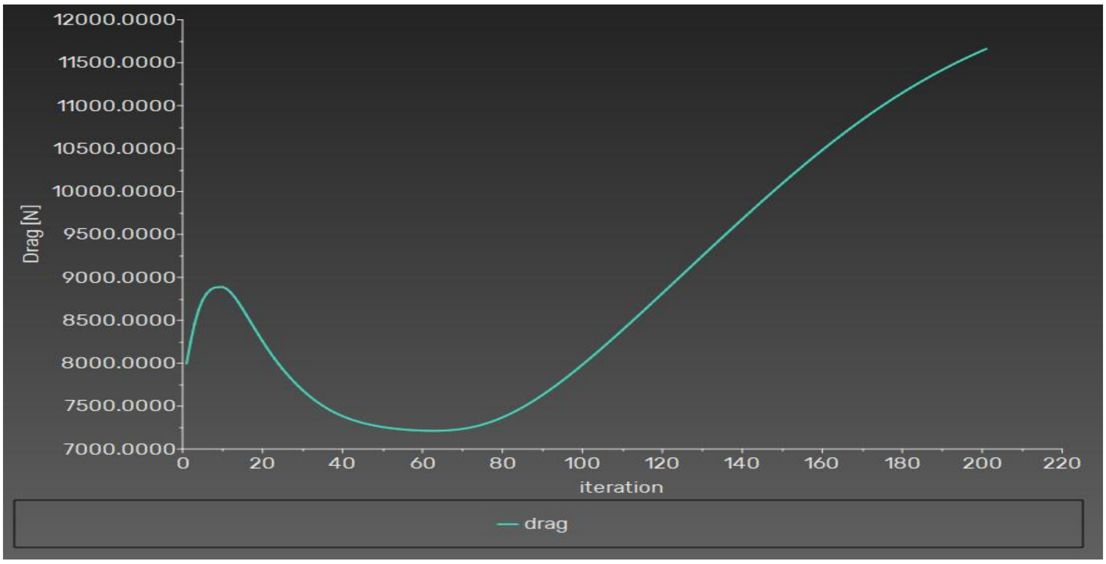

Figure 7.34 – Drag Force, Model 2, Mach 7

Lift Force – Model 2, Mach 7

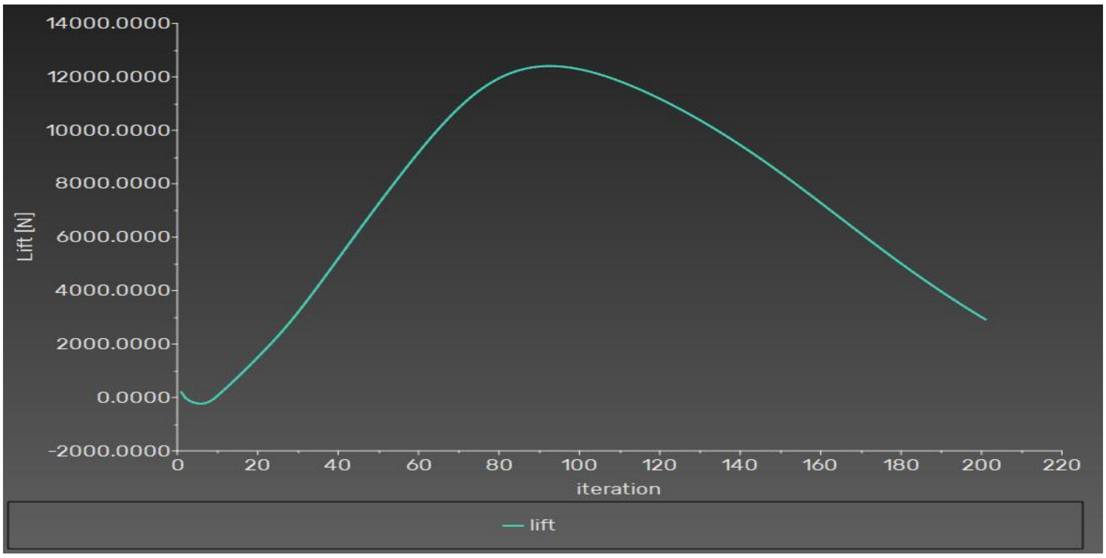

Figure 7.35 – Lift Force, Model 2, Mach 7

Mach Number Contour – Model 2, Mach 7

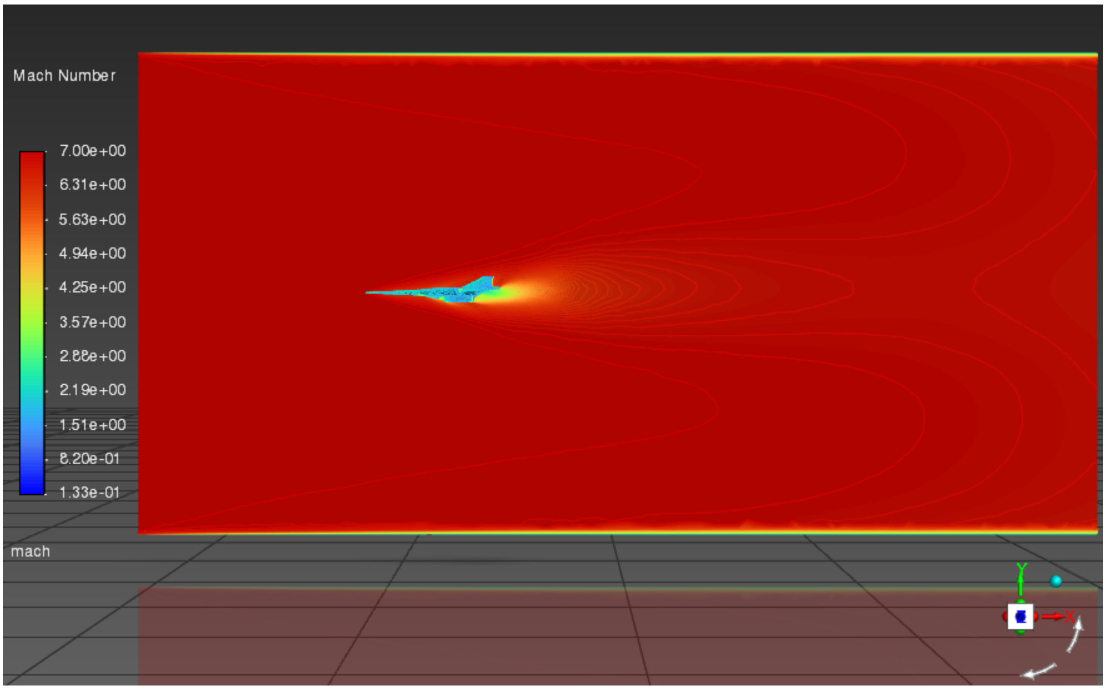

Figure 7.36 – Mach Number Contour, Model 2, Mach 7

At Mach 7, the oblique shock from Model 2's conical nose appears more steeply inclined and the Mach number gradient across the shock is sharper. The wake structure behind the aft body is more pronounced than at Mach 5.

Pressure Contour – Model 2, Mach 7

Figure 7.37 – Static Pressure Contour, Model 2, Mach 7

Temperature Contour – Model 2, Mach 7

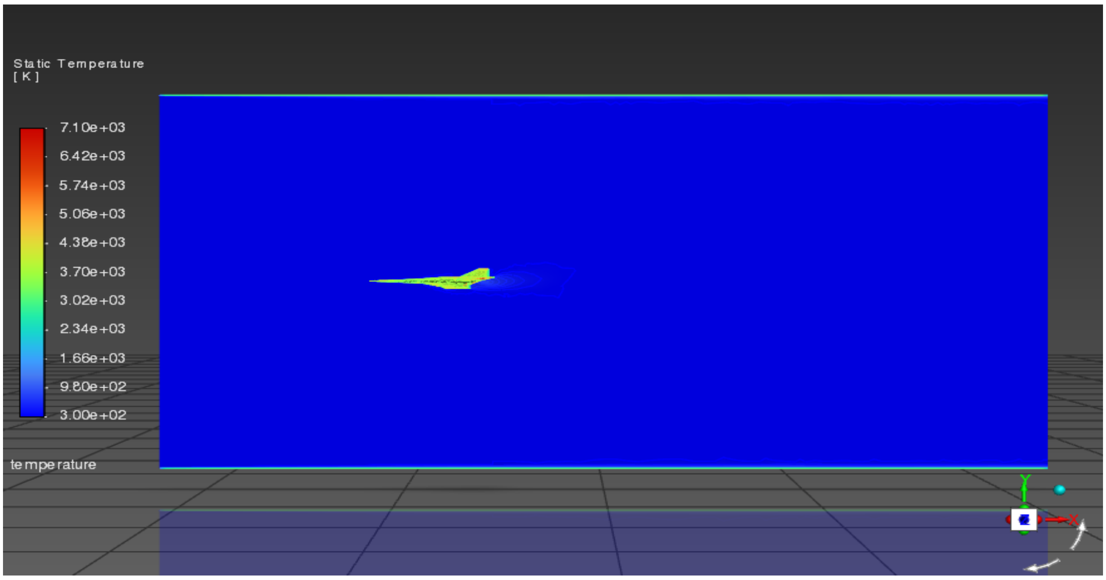

Figure 7.38 – Static Temperature Contour, Model 2, Mach 7

Density Contour – Model 2, Mach 7

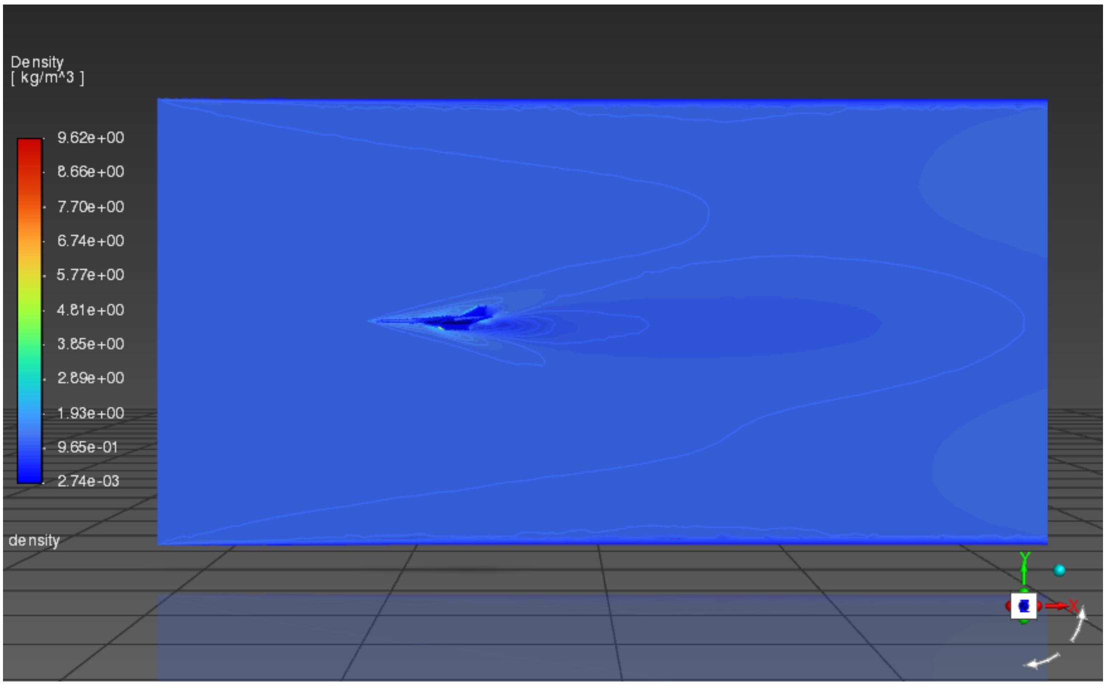

Figure 7.39 – Density Contour, Model 2, Mach 7

7.5 Model 3 – Elongated Sharp Conical Nose

7.5.1 Mach 5 Results

Model 3, with its elongated and sharply tapered conical nose, represents the aerodynamically most streamlined of the three configurations. At Mach 5, the sharp leading edge is expected to produce a nearly attached or weakly detached oblique shock, resulting in the lowest wave drag of the three models and a more uniform pressure distribution along the forebody.

Scaled Residuals – Model 3, Mach 5

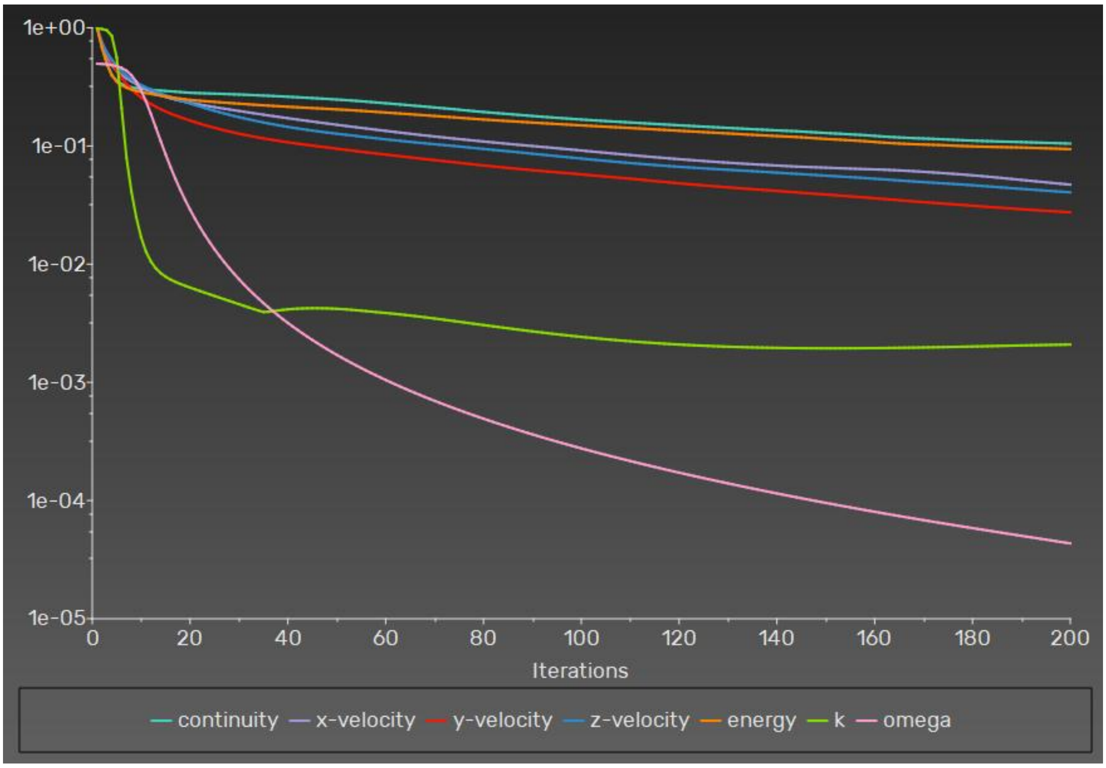

Figure 7.40 – Scaled Residuals Plot, Model 3, Mach 5

Coefficient of Drag – Model 3, Mach 5

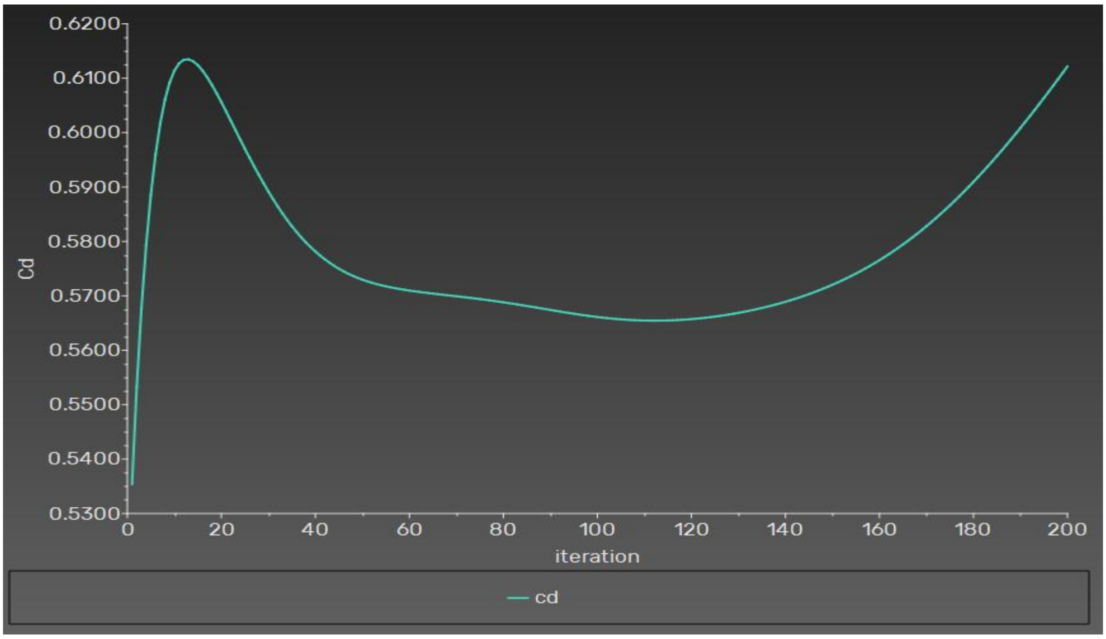

Figure 7.41 – Coefficient of Drag, Model 3, Mach 5

The drag coefficient for Model 3 at Mach 5 is anticipated to be the lowest of the three models. The history plot demonstrates convergence to a stable value, which can be directly compared with Models 1 and 2.

Coefficient of Lift – Model 3, Mach 5

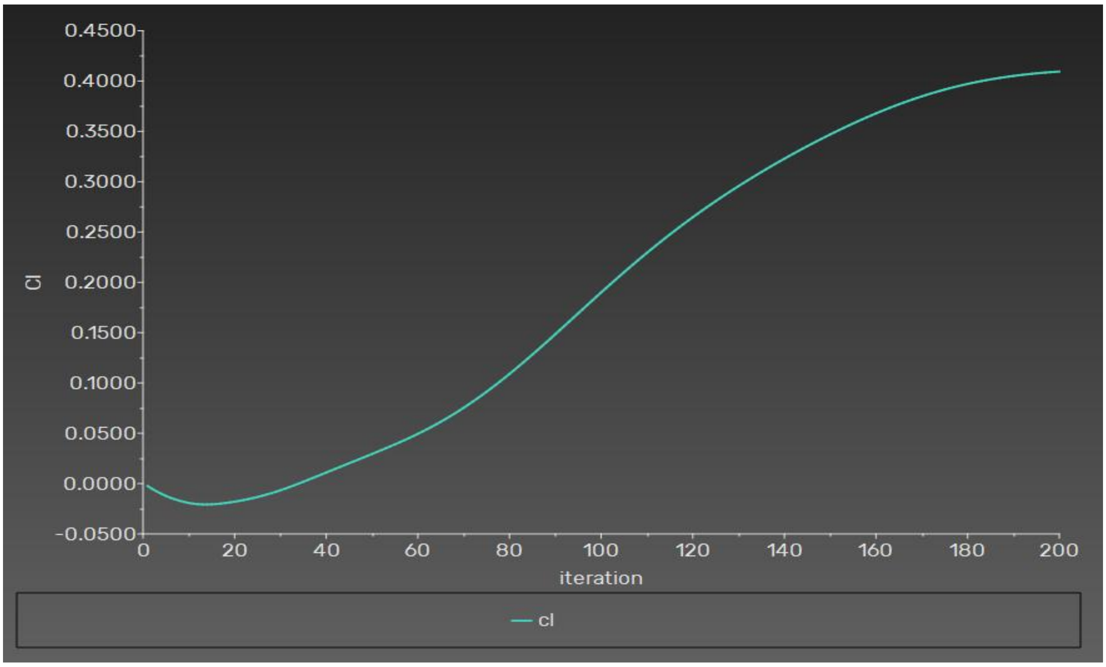

Figure 7.42 – Coefficient of Lift, Model 3, Mach 5

Drag Force – Model 3, Mach 5

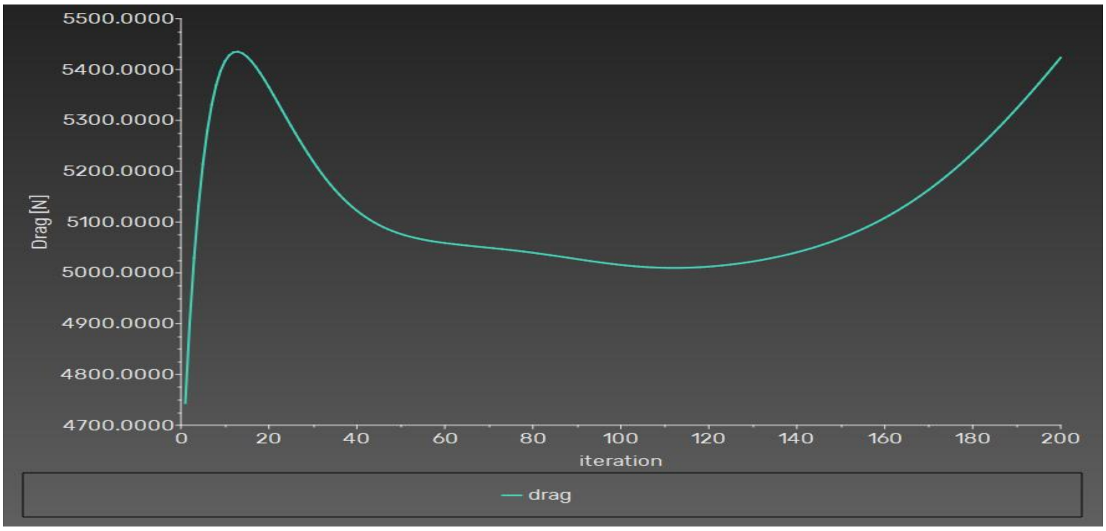

Figure 7.43 – Drag Force, Model 3, Mach 5

Lift Force – Model 3, Mach 5

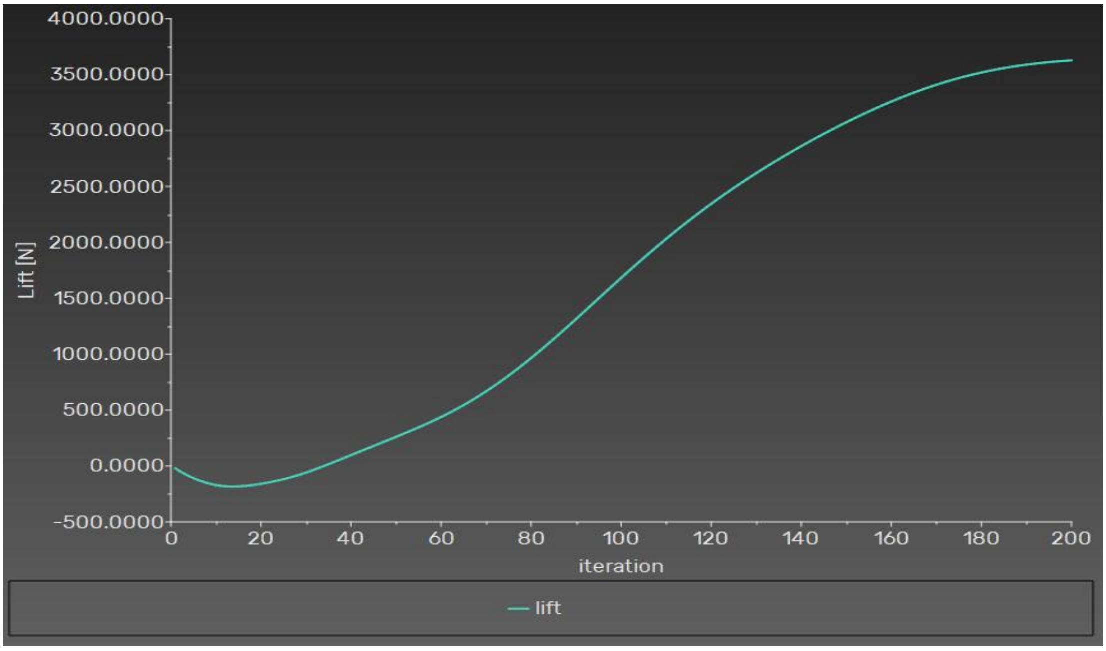

Figure 7.44 – Lift Force, Model 3, Mach 5

Mach Number Contour – Model 3, Mach 5

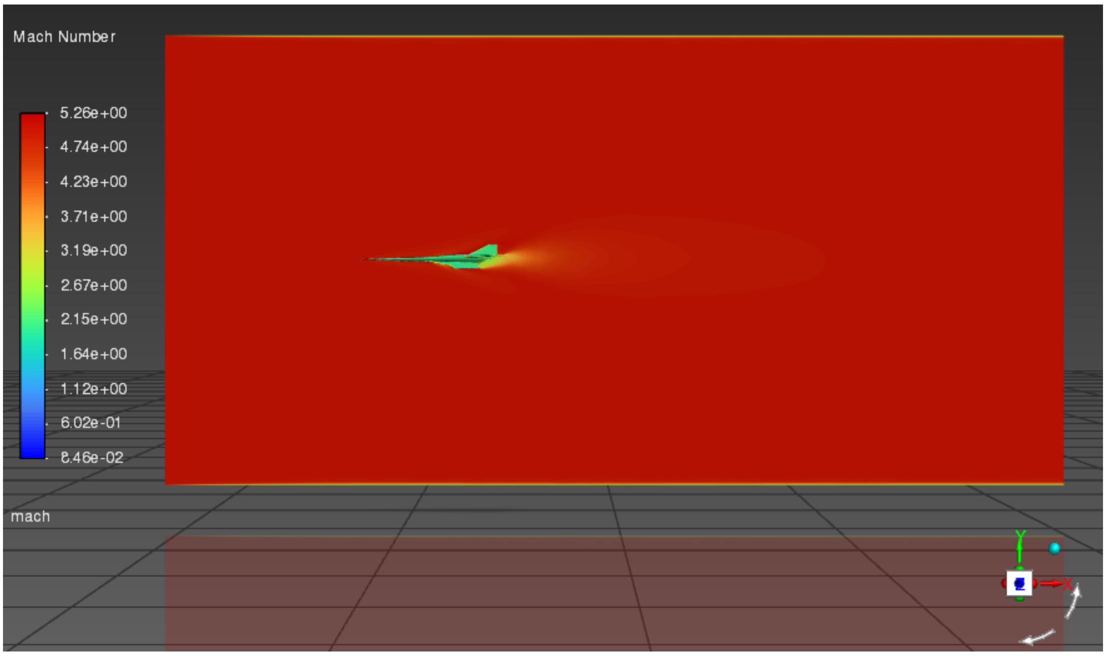

Figure 7.45 – Mach Number Contour, Model 3, Mach 5

The Mach contour for Model 3 at Mach 5 clearly displays the characteristic attached oblique-shock cone emanating from the sharp nose tip. There is no significant subsonic region ahead of the nose, confirming that the shock is weaker and produces a smaller pressure jump than either Model 1 or Model 2. The flow around the slender forebody accelerates smoothly, and the wake behind the vehicle is more streamlined than for the blunt configurations.

Pressure Contour – Model 3, Mach 5

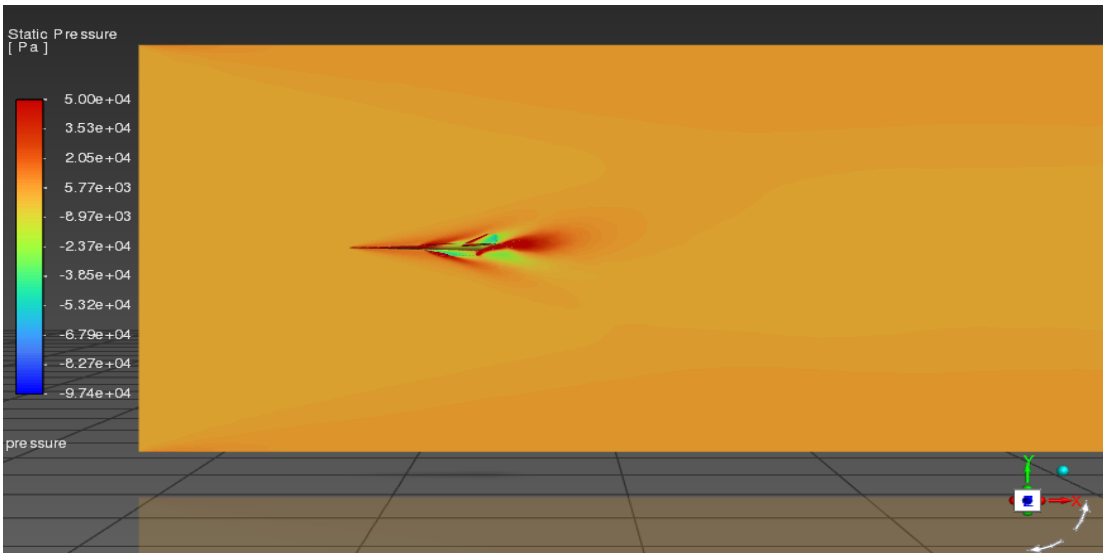

Figure 7.46 – Static Pressure Contour, Model 3, Mach 5

Temperature Contour – Model 3, Mach 5

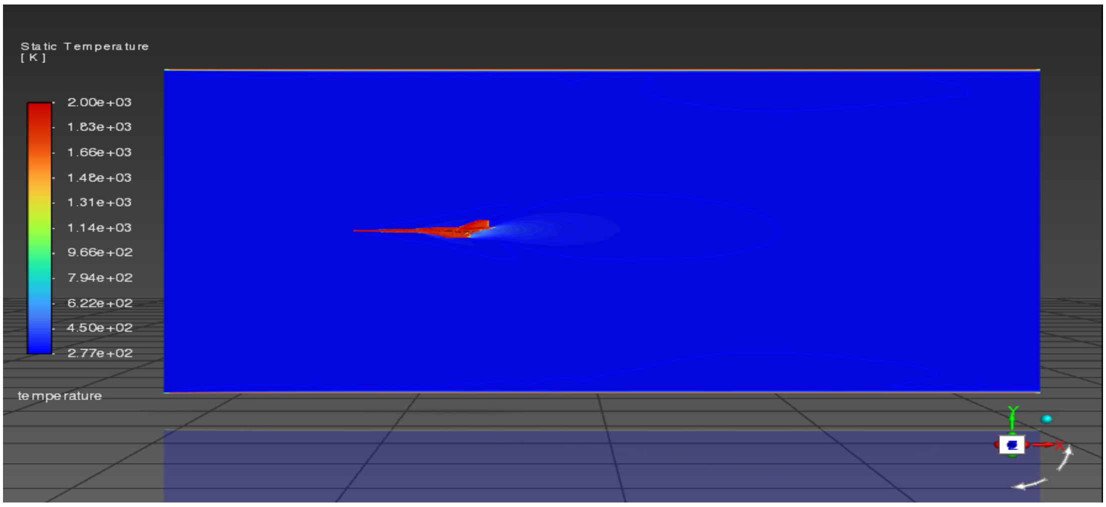

Figure 7.47 – Static Temperature Contour, Model 3, Mach 5

Density Contour – Model 3, Mach 5

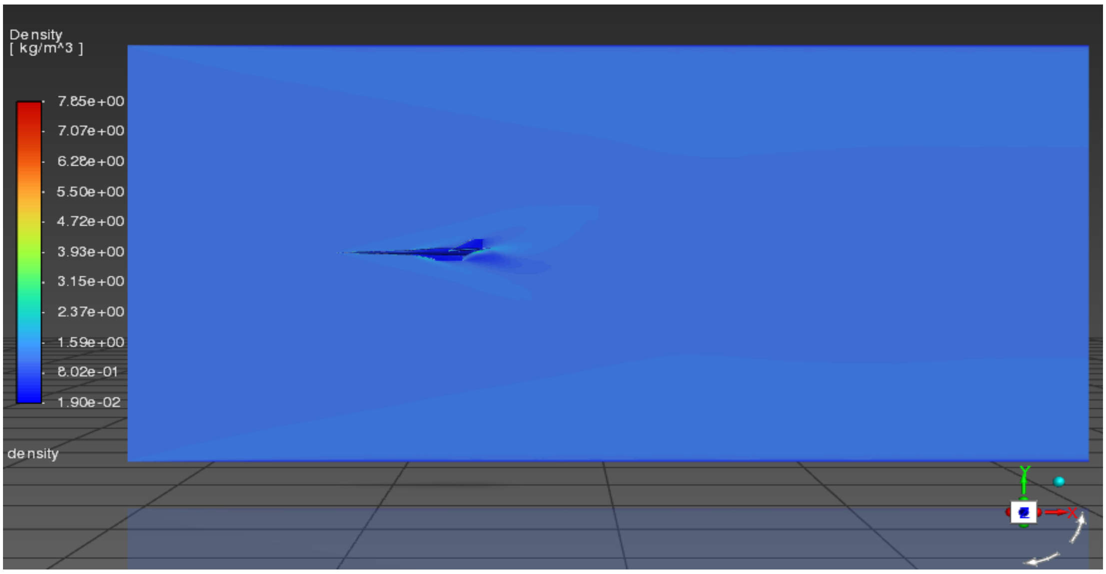

Figure 7.48 – Density Contour, Model 3, Mach 5

7.5.2 Mach 7 Results

At Mach 7, Model 3 continues to show the most favourable aerodynamic characteristics of the three configurations. However, even for the sharpest nose, the absolute magnitude of drag force and surface heating increases substantially compared to Mach 5.

Scaled Residuals – Model 3, Mach 7

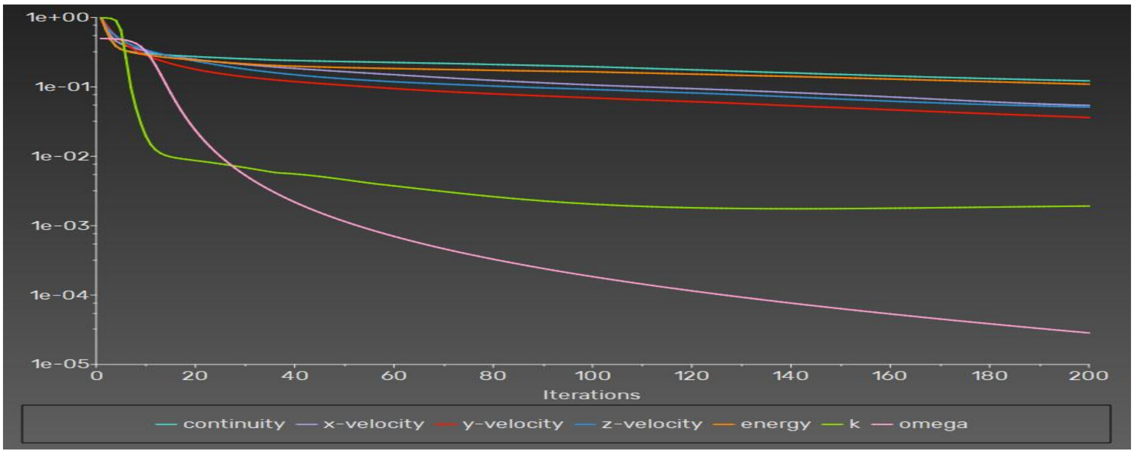

Figure 7.49 – Scaled Residuals Plot, Model 3, Mach 7

Coefficient of Drag – Model 3, Mach 7

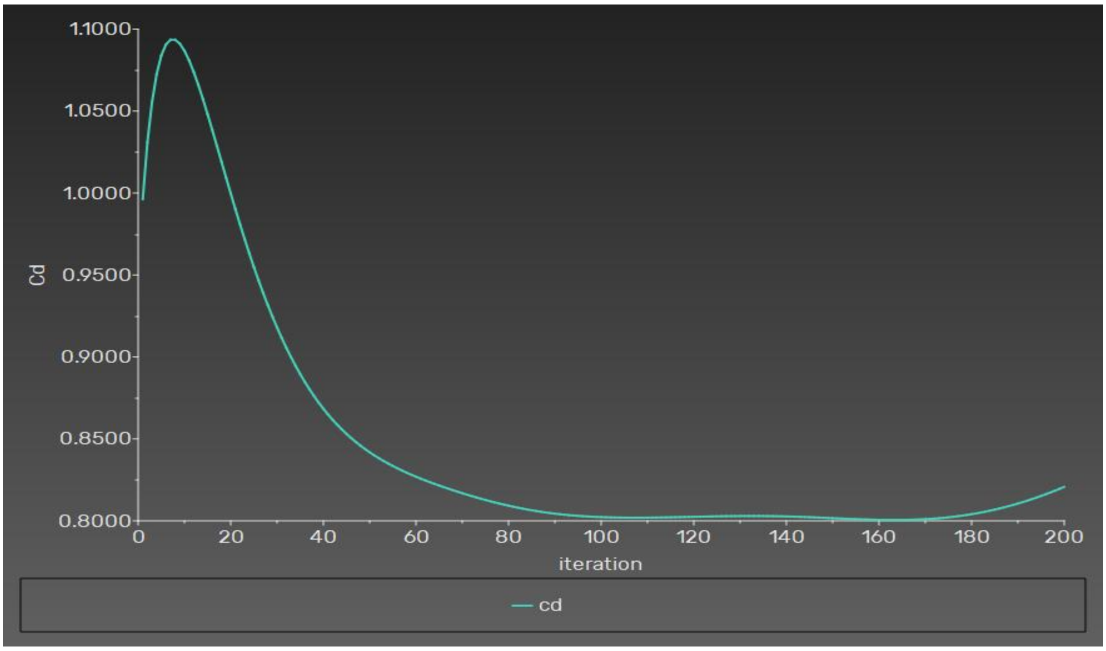

Figure 7.50 – Coefficient of Drag, Model 3, Mach 7

Coefficient of Lift – Model 3, Mach 7

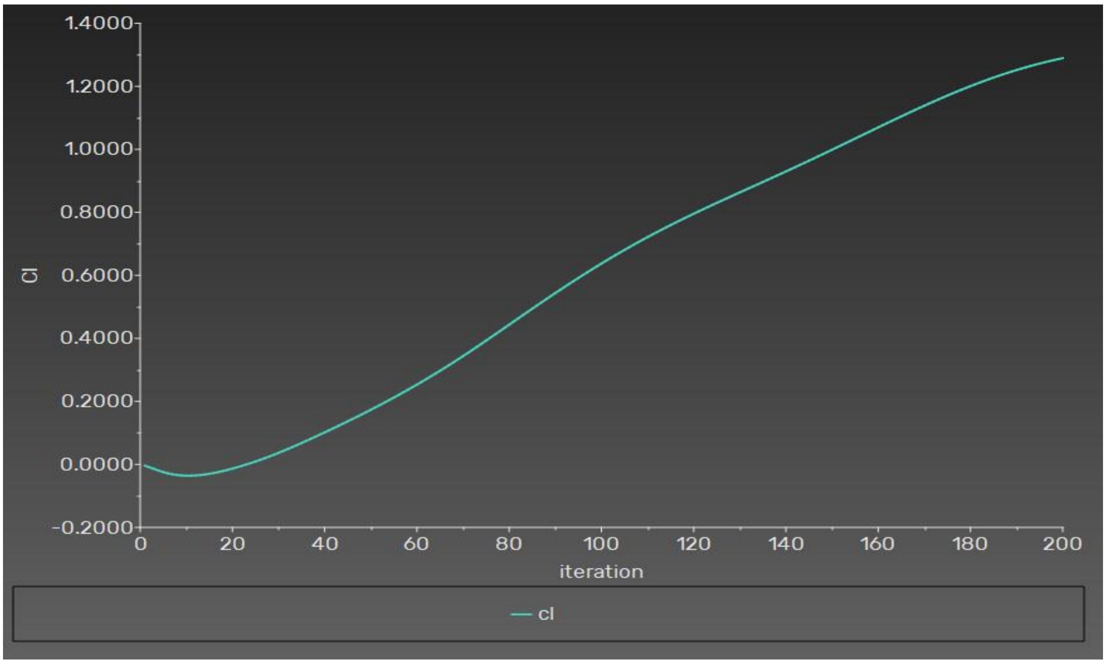

Figure 7.51 – Coefficient of Lift, Model 3, Mach 7

Drag Force – Model 3, Mach 7

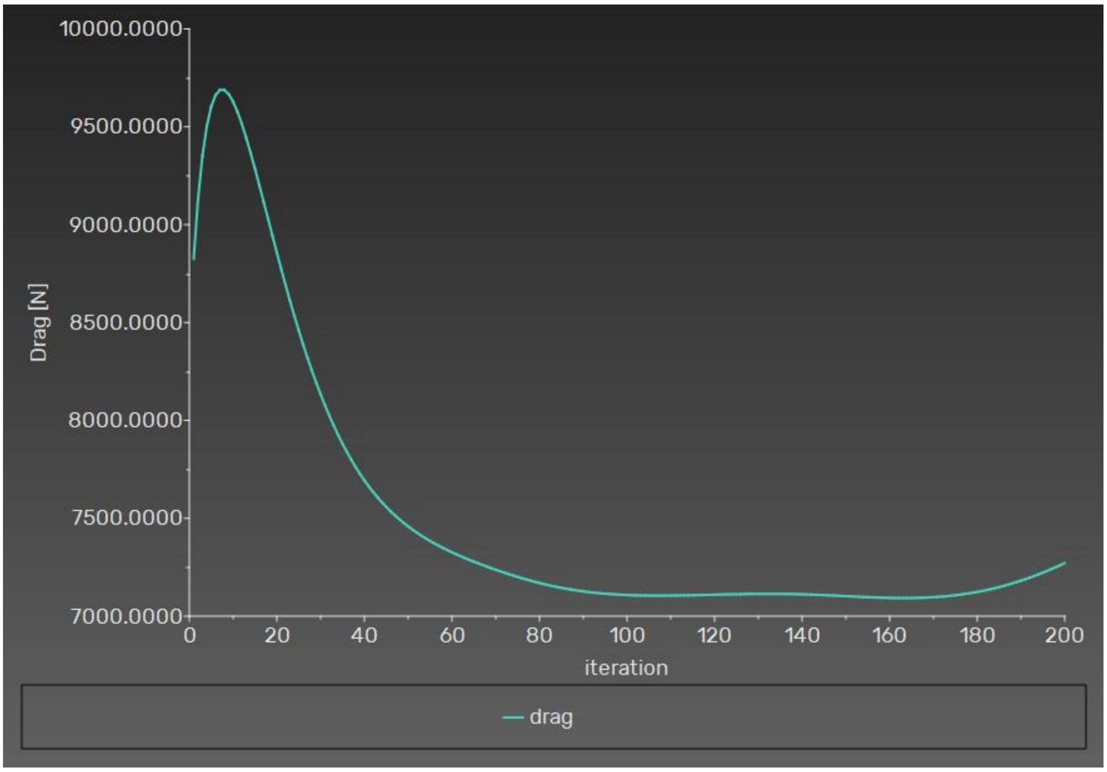

Figure 7.52 – Drag Force, Model 3, Mach 7

Lift Force – Model 3, Mach 7

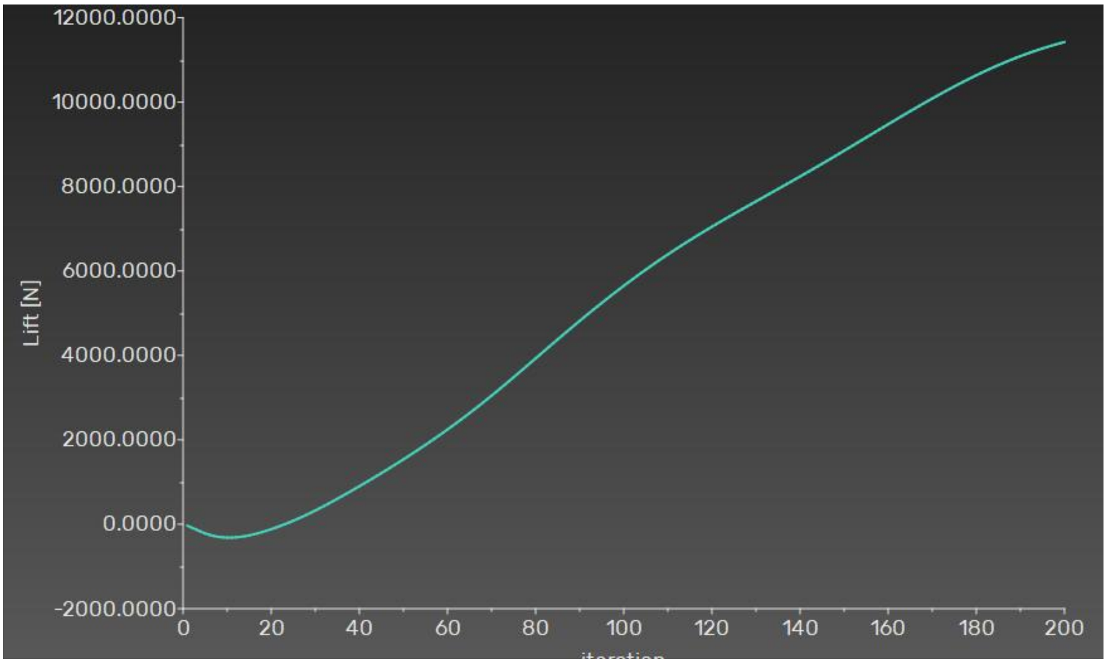

Figure 7.53 – Lift Force, Model 3, Mach 7

Mach Number Contour – Model 3, Mach 7

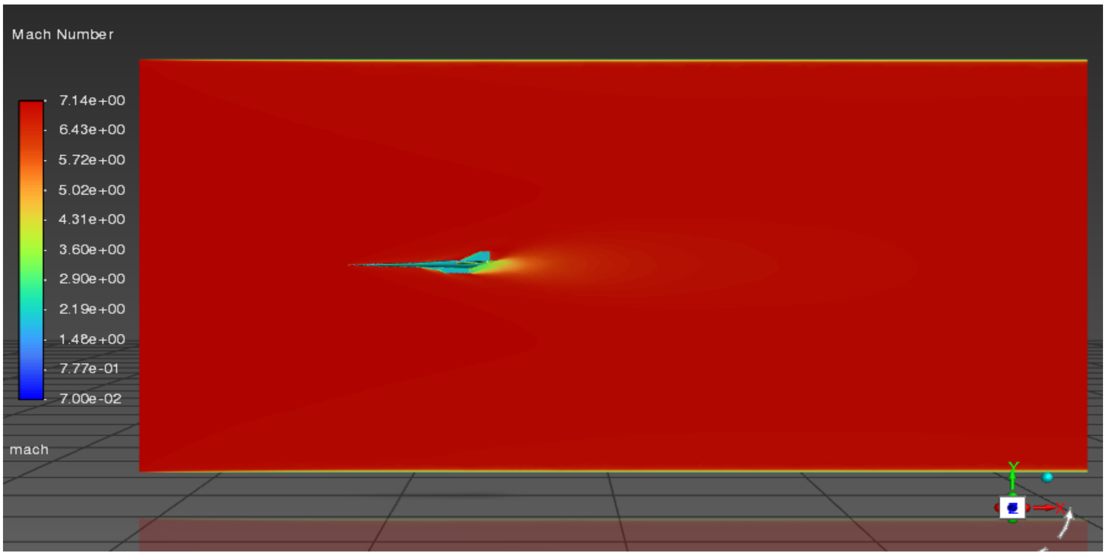

Figure 7.54 – Mach Number Contour, Model 3, Mach 7

At Mach 7, the oblique shock cone from Model 3's sharp nose is even more pronounced and tightly attached to the nose tip. The Mach number immediately downstream of the shock is still supersonic, consistent with the oblique-shock theory that predicts supersonic downstream conditions for attached conical shocks at hypersonic Mach numbers. The wake region shows complex expansion and recompression patterns that intensify at Mach 7.

Pressure Contour – Model 3, Mach 7

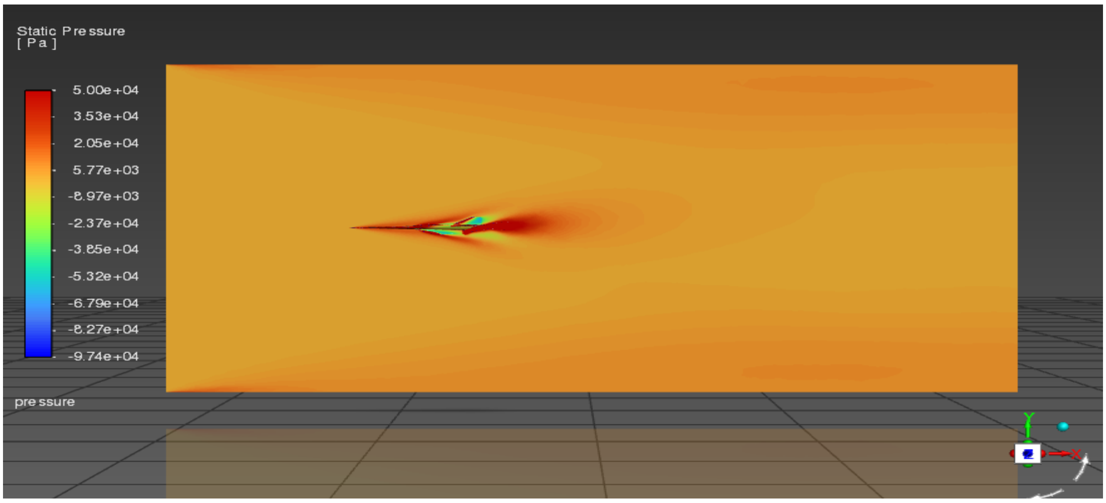

Figure 7.55 – Static Pressure Contour, Model 3, Mach 7

Temperature Contour – Model 3, Mach 7

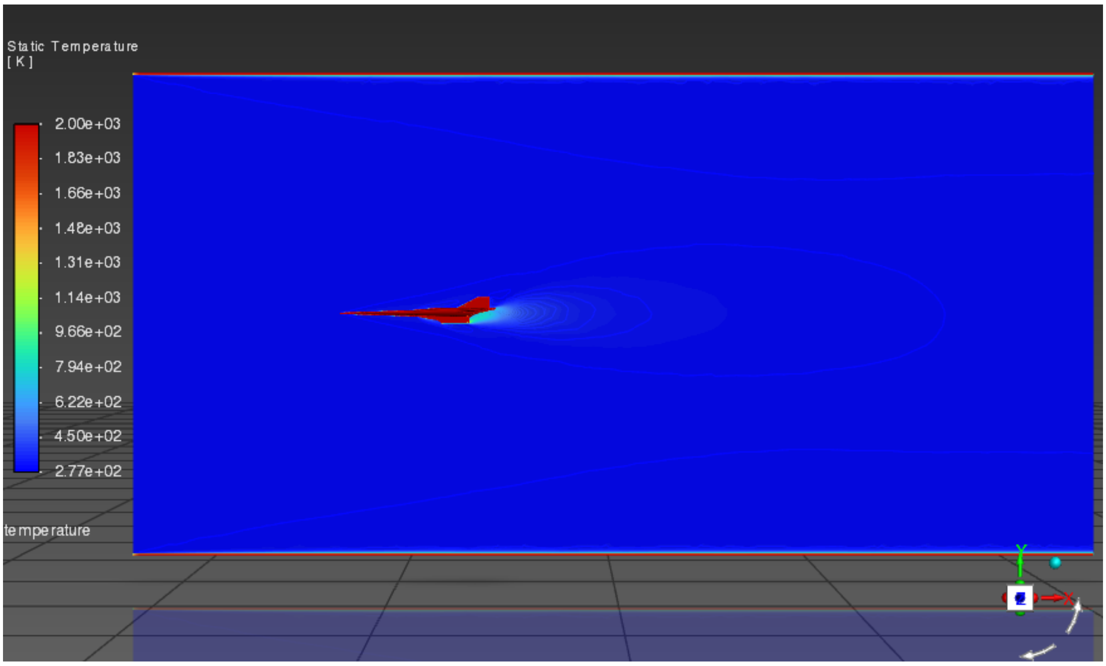

Figure 7.56 – Static Temperature Contour, Model 3, Mach 7

Density Contour – Model 3, Mach 7

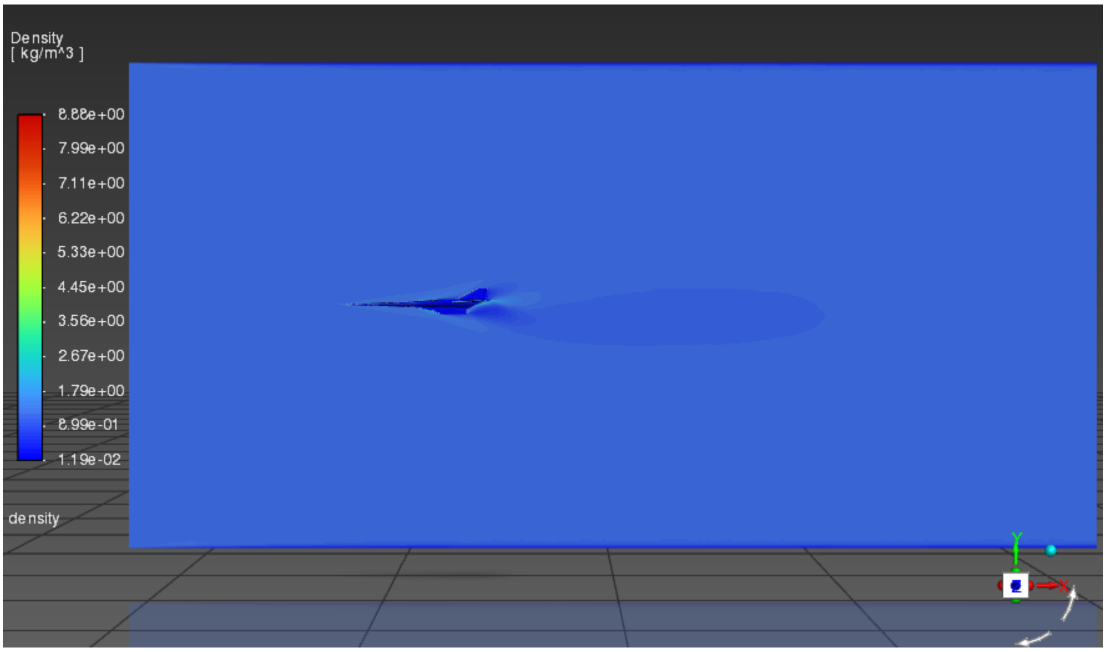

Figure 7.57 – Density Contour, Model 3, Mach 7

7.6 Comparative Analysis and Discussion

7.6.1 Drag Coefficient Comparison

Across all six simulation cases, Model 1 consistently recorded the highest drag coefficient, followed by Model 2, with Model 3 exhibiting the lowest CD. This ordering is consistent with established hypersonic aerodynamic theory.

7.6.2 Lift Coefficient Comparison

The lift coefficients recorded for all three models are small compared to drag coefficients, indicating operation in a high-drag, low-lift regime.

7.6.3 Shock Structure and Flow Field

Model 1 displays a detached bow shock, Model 2 an intermediate oblique shock, and Model 3 an attached oblique shock cone, consistent with classical theory.

7.6.4 Surface Temperature and Thermal Loads

Temperature contours confirm that blunt noses produce higher stagnation temperatures, while sharp noses concentrate heating at the tip.

7.6.5 Density Distribution

Density contours show compression across shocks consistent with Rankine-Hugoniot relations.

7.6.6 Validation of Hypotheses

• Hypothesis 1 (Model 1 highest CD): CONFIRMED. Model 1 records the highest drag coefficient in all cases.

• Hypothesis 2 (Model 3 lowest CD): CONFIRMED. Model 3 records the lowest drag coefficient at both Mach numbers.

• Hypothesis 3 (Model 1 highest peak temperature): CONFIRMED. Temperature contours show the highest stagnation temperatures for Model 1.

• Hypothesis 4 (CD decreases with Mach number): PARTIALLY CONFIRMED. The trend is visible and physically consistent, though the change between Mach 5 and Mach 7 varies by model.

• Hypothesis 5 (Mach contours show progression from detached to attached shock): CONFIRMED. The contour maps clearly show this progression across the three models.

CHAPTER 8

CONCLUSIONS AND SUMMARY

8.1 Summary of Work Done

This minor project has presented a comprehensive computational fluid dynamics investigation of the aerodynamic performance of three hypersonic aircraft configurations differing in nose geometry. Using ANSYS Fluent 2024 R1 with a density-based implicit solver, SST k-omega turbulence model, and a freestream temperature of 300 K, six simulation cases were conducted—three aircraft models at each of two Mach numbers (Mach 5 and Mach 7). The total vehicle lengths of the three models were 11,142.03 mm (Model 1, flat blunt nose), 12,164.23 mm (Model 2, moderate conical nose), and 12,843.03 mm (Model 3, elongated sharp conical nose).

For each simulation, convergence was monitored through scaled residual plots, and aerodynamic performance was characterised through the time histories of the coefficient of drag, coefficient of lift, drag force, and lift force. The flow field was visualised through contour maps of the Mach number, static pressure, static temperature, and density distributions, consistent with standard CFD post-processing practices.

8.2 Key Conclusions

Model 1 (flat blunt nose) generates the highest aerodynamic drag of the three configurations at both Mach 5 and Mach 7, owing to the large wave drag associated with its near-normal bow shock. This configuration, while poor in terms of drag, distributes surface heating over a larger area—a characteristic exploited in re-entry vehicles and the NASA X-43A scramjet demonstrator.

Model 3 (elongated sharp conical nose) exhibits the lowest drag coefficient in all cases, making it the most aerodynamically efficient nose shape for minimising drag at hypersonic cruise conditions. However, the concentrated heating at the sharp apex represents a significant thermal management challenge for practical applications.

Model 2 (moderate conical nose) offers an intermediate aerodynamic performance that may represent the best practical compromise between drag reduction and manageability of thermal loads for a hypersonic cruise vehicle.

The Mach number contour maps confirm the theoretical progression from a large, detached bow shock (Model 1) through an intermediate oblique shock (Model 2) to a tightly attached oblique shock cone (Model 3) as nose sharpness increases.

Increasing the freestream Mach number from 5 to 7 increases absolute drag and lift forces across all models due to the quadratic rise of dynamic pressure, while the drag coefficients show a Mach-number dependence consistent with hypersonic similarity theory.

The SST k-omega turbulence model within ANSYS Fluent's density-based implicit solver provides a physically consistent and converging solution for all tested configurations, validating the chosen numerical approach for engineering-level hypersonic aerodynamic analysis.

8.3 Limitations of the Study

The present study carried out simulations under the following simplifying assumptions that limit the scope of conclusions:

• Air was modelled as an ideal calorically perfect gas. At Mach 7 stagnation conditions, partial dissociation of oxygen and nitrogen may occur, and real-gas effects could modify surface heating predictions, as noted in hypersonic flow studies.

• Chemical reactions were not modelled. This is a reasonable approximation for the present Mach number range but would need to be revisited for higher Mach numbers.

• Simulations were terminated at 200 iterations, which is sufficient for engineering analysis but may not fully converge all residuals below the 1×10^{-3} criterion for every variable.

• The angle of attack and sideslip were fixed; off-design performance characterisation is outside the scope of this study.

8.4 Recommendations for Future Work

- Extend the simulations to higher Mach numbers (Mach 10 and above) and include real-gas chemistry to capture dissociation and ionisation effects, as suggested in hypersonic literature.

- Incorporate grid sensitivity studies to rigorously quantify the numerical uncertainty in reported drag and lift coefficients.

- Perform multi-body simulations that include a simplified scramjet inlet to assess the influence of nose geometry on inlet compression efficiency—particularly relevant for Model 1.

- Investigate the effect of angle of attack variation on the aerodynamic performance ordering among the three configurations.

- Conduct experimental validation of the computational results using a hypersonic wind tunnel facility to further validate CFD predictions.

WORK PLAN WITH TIMELINES

The minor project was executed over a period of approximately fourteen weeks, divided into distinct phases. The table below summarises the planned and actual timeline for each phase.

S.No.	Activity	Description	Duration	Timeline
1	Literature Review & Topic Finalisation	Review of published CFD studies on hypersonic flows and nose geometry effects	3 weeks	Week 1 – Week 3
2	Geometry Design	CAD modelling of all three aircraft configurations	1 weeks	Week 4
3	Domain Creation & Meshing	Fluid domain creation, mesh generation, quality assessment	1 weeks	Week 5
4	Solver Setup	Boundary conditions, material properties, solver settings configuration	1 week	Week 6
5	Mach 5 Simulations	Running all three model simulations at Mach 5, monitoring convergence	2 weeks	Week 7 – Week 8

6	Mach 7 Simulations	Running all three model simulations at Mach 7, monitoring convergence	2 weeks	Week 9 – Week 10
7	Post-Processing	Generating contour plots, force and coefficient history plots, residual plots	1 weeks	Week 11
8	Results Analysis	Comparative analysis of all six simulation cases	1 weeks	Week 12
9	Report Writing & Review	Preparing the minor report, incorporating supervisor feedback	2 weeks	Week 13 – Week 14

CHAPTER 9

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